University of Strathclyde Engineering

PATH PLANNING FOR ULTRASONIC INSPECTION OF SADDLE SHAPED WELDS USING LASER AND TACTILE FEEDBACK

Jakub Dysko, Adam Gilmour, Gordon Dobie

Abstract

Due to constantly increasing standards for safety and efficiency for quality inspection of industrial components, there is great demand for automation within this field. Many modern inspection systems still involve a large degree of human intervention for conducting non-destructive testing (NDT) of pipe welds which leads to workers operating in unsafe environments, and risks human error compromising inspection standards. This body of work reviews the current NDT state of the art and outlines various examples of relevant work, including robotic systems with multi-sensor feedback. A system is proposed as the groundwork for a fully autonomous inspection device for saddle-shaped welds which is capable of extracting the radii and position of two intersecting pipes using laser-based point cloud acquisition and the RANSAC algorithm, and planning an ultrasonic (US) inspection path for the saddle-shaped intersection curve within the robots working range. A force-torque sensor is used to mitigate inaccuracies in the planned path during the US scan. The sub-components of the system are evaluated with various tests and opportunities for future work are highlighted.

Keywords: Automated NDE/NDT, Robotic NDE/NDT, Robotic Path Planning, Point Cloud Navigation, Force-Torque Feedback, Phased Array Ultrasonic Inspection, Complex Geometry

SEPTEMBER 3, 2025
UNIVERSITY OF STRATHCLYDE
16 Richmond St, Glasgow G1 1XQ

Contents

Abstract 0
Contents 1
Table of figures 2
Table of tables 3
1 Introduction 3
1.1 Background 3
1.1.1 NDT/NDE overview 3
1.1.2 NDT systems 6
1.2 Review of related work 8
1.2.1 Software based trajectory planning 9
1.2.2 Visual based trajectory planning 10
1.2.3 Force-torque sensor integration 11
1.2.4 Trajectory planning using ultrasonic feedback 12
1.2.5 Saddle path trajectory planning 12
1.3 Review of NDT systems at Strathclyde 13
1.3.1 System 1 – enhancing positional awareness 13
1.3.2 System 2 – combination of RGB camera data and force-torque sensor feedback 14
1.3.3 System 3 – improving probe positioning using ultrasonic imaging 15
1.3.4 System 4 – multi-sensor robotic inspection system 15
2 Scope of work 16
3 System overview 17
3.1 Physical setup 17
3.2 System operation 20
3.2.1 Point cloud generation 20
3.2.2 Path generation 23
3.2.3 Path execution 31
4 Testing 33
4.1 Laser scanning 33
4.2 Force-torque feedback 36
4.3 Overall system 37
5 Further work 37
6 Conclusion 39
7 Appendices 40

7.1 Appendix A: Mock pipe section 3D drawing 40
7.2 Appendix B: Mount 3D drawing 41
8 References 41

Table of figures

Figure 1 Ultrasonic test basic visual description. The pulse fired by the transducer is partially reflected by one of the voids causing a minor pulse to be observed prior to the main reflection as seen on the right 5
Figure 2 Phased array ultrasonic wave construction. On the left, the different probes with their individual time delays can be seen, with the overall resultant waves and their direction shown on the right 5
Figure 3 Tracked wall-climbing robot for ship inspection. Possesses magnetic tracks as shown and is equipped with an ultrasonic probe for NDT. The device is also compatible with other NDT instruments 6
Figure 4 Volio Tricopter diagram. This diagram visualises the robot's tiltable blades and includes a surface which is making contact with the robot end effector 7
Figure 5 UUV prototype. Possesses a neutrally buoyant rigid body and a series of phased array probes for conducting NDT 7
Figure 6 Pipe saddle weld 3D model 8
Figure 7 NDE system with force-torque sensor for additional feedback 11
Figure 8 Variation in cross section of saddle weld with respect to angular position about smaller pipe. 12
Figure 9 Meca500 arm mobile NDT system application. As can be seen the robot can maintain adhesion on a pipe surface 13
Figure 10 Constructed image of weld internal cross section using ultrasonic tests. The weld is invisible to the naked eye but its shape is clearly seen in the figure 14
Figure 11 Generated waypoints for raster scan on curved surface [40] 15
Figure 12 Remote inspection system where the base carries a small robotic manipulator with an ultrasonic payload 16
Figure 13 Meca500 robotic arm and power box used to carry out work 17
Figure 14 (a) CAD model of mock inspection unit with pipe section and Meca500 arm equipped with ultrasonic probe wedge and FT sensor (b) real-life setup with dummy probe equipped 17
Figure 15 System set up on real life pipe sample using magnetic roller probe 18
Figure 16 CAD model of mount for phased array probe and Opto1420 laser 18
Figure 17 Process summary 19
Figure 18 Mount with laser and wedge attached showing laser point offset from TRF. Note that dX, dY and dZ do not correspond to X, Y and Z offsets due to the TRF having a different orientation 20
Figure 19 Example of generated point cloud from laser scan displayed using PyVista library 22
Figure 20 Scan paths 1 and 2 visualised on pipe intersection 22
Figure 21 Local coordinate frame at intersection of cylindrical axes visualised 23

Figure 22 (a) YZ plane view of pipe intersection showing required roll angle for z axis to be anti-normal to the horizontal pipe surface. (b) shows the required yaw angle for y axis to be anti-normal to vertical pipe surface. Roll operation must be prior to yaw.....24
Figure 23 Saddle curved mapped onto 2D plane visualised. This curve describes how the z value changes with respect to position along the circumference of the vertical pipe.....25
Figure 24 (a) TRF path for saddle inspection 3D view (b) YZ plane view showing tangency to weld and anti-normality to horizontal pipe surface (c) XY plane view showing anti-normality to vertical pipe.....26
Figure 25 Generated point cloud data from laser scan with RANSAC fitted cylinders.....27
Figure 26 Fitted cylinders with local coordinate frames visualised.....28
Figure 27 (a) Generated TRF path with offset included ensuring constant offset from fitted vertical cylinder. (b) offset omitted showing inaccurate positioning of scan path 1 for erroneous cylinder fit.....29
Figure 28 Point cloud with cylinders fitted and scan paths generated.....30
Figure 29 Flowchart for PID control system.....31
Figure 30 Wooden block for laser test.....32
Figure 31 Generated point cloud from laser scan on wooden block showing approximate 43 x 43 mm front face.....32
Figure 32 MATLAB cylinder fit on point cloud image generated using Zivid camera.....33
Figure 33 CloudCompare cylinder fit on point cloud image generated using Zivid camera..33
Figure 34 (a) two scans (base positioned centrally), slow speed (b) four scans (base positioned to the left, then to the right), slow (c) four scans, medium (d) four scans, fast (e) four scans, slow, axial rather than circumferential.....34
Figure 35 (a) system response with extreme gain causing significant oscillations (b) system response with moderate gain resulting in minor oscillations (c) system response with finely tuned gain showing convergence.....35
Figure 36 (a) force response from optimised PID parameters (b) torque responses from optimised PID parameters.....36

Table of tables

Table 1 Comparison between DT and NDT.....4
Table 2 Joint parameters for laser scans.....21
Table 3 Results from laser scanning tests.....34

1 Introduction

1.1 Background

1.1.1 NDT/NDE overview

The strive for safety and efficiency within the industrial sector is at the forefront of modern engineering [1]. In particular, quality inspection of geometrically complex welds within infrastructure is becoming increasingly automated thanks to constantly advancing robotic and sensing technology [2]. Until now, inspection methods have required a large degree of human involvement, giving rise to safety concerns, human error, time and data collection

constraints, and more [3]. Industrial component inspections are typically undergone using non-destructive testing and evaluation (NDT/NDE), which are methods that can be performed directly on components without causing damage to the part under inspection. There are a wide range of NDT methods, such as Ultrasonic Testing (UT), Magnetic Particle Testing, Eddy Current Testing, Microwave and Millimeter Wave Testing, Radiography Testing, Penetrant Testing, Acoustic Emission Testing and Thermography [4]. NDT has various advantages over destructive testing (DT), summarised by table 1 [5] which includes the advantages and limitations of both methods:

Table 1 Comparison between DT and NDT

Destructive tests	Non-destructive tests
Advantages	**Limitations**
1. Measurements are direct and reliable	Measurements are indirect reliability is to be verified
2. Usually quantitative measurements	Usually qualitative measurements
3. Correlation between test measurements and material properties are direct	Measurements can also be done quantitatively. Skilled judgement and experience are required to interpret indications
Limitations	**Advantages**
1. Tests are not made on the objects directly. Hence correlation between the sample specimen used and object needs to be proved.	Tests are made directly on the object. 100% testing on actual components is possible.
2. A single test may measure only one or a few of the properties.	Many NDT methods can be applied on the same part and hence many or all properties of interest can be measured.
3. Inservice testing is not possible	Inservice testing is possible
4. Measurement of properties over a cumulative period of time cannot readily be possible	Repeated checks over a period of time are possible.
5. Preparation of the test specimen is costly	Very little preparation is sufficient
6. Time requirements are generally high	Most test methods are rapid

NDT has developed since the turn of the 20th century, with rapid advancements particularly after the second world war [6]. Before any form of equipment was available, humans would rely on their own senses and perceptions to inspect components for defects and fitness for purpose. The development of procedures, using enablers such as soot, coal, and chalk, allowed more rigorous and detailed testing to be conducted [7]. Chemical and physical phenomena, such as ultrasonic and electromagnetic waves, then began to be utilised as a means of observing internal characteristics of components. Many modern NDT methods incorporate autonomous elements with the emergence of robotic systems.

A revolution is currently taking place, often described as industry 4.0 or the 4th industrial revolution, characterised by informatisation, digitalisation, networks, interfaces, digital communication, artificial intelligence, machine learning, 5G and quantum technologies [8]. Similar developments are taking place specifically within the NDE industry, whereby the characteristics of industry 4.0 are being incorporated to further autonomise and improve NDT procedures. Collected data will be used to create databases and markets of information which can be networked between varying telecommunication technologies.

One of the industry's most significant developments in the 20th century was the advent of ultrasonic NDT [6]. This method is of particular focus within this field of work and will be used as a means of inspection for a pipe saddle weld. Ultrasonic NDT was selected due to its ability to conduct volumetric inspections in metallic materials.

Ultrasonic NDE is conducted by firing pulses into a material and observing the response from these pulses. As an ultrasonic pulse reaches the boundary of two media, a proportion of the wave's energy is reflected. If a pulse is fired into a component, a reflection will occur as a result of the wave reaching the outer surface of the component. Cracks within the component will cause small proportions of the wave energy to be reflected early, hence observation of reflected pulses can give rise to information describing the internal state of the component. An ultrasonic pulse is created and received using a piezoelectric transducer, composed of a material which converts electrical current into mechanical expansion and contraction, and vice versa. Figure 1 [4] provides a basic visual representation of an ultrasonic test on a metal block containing internal defects:

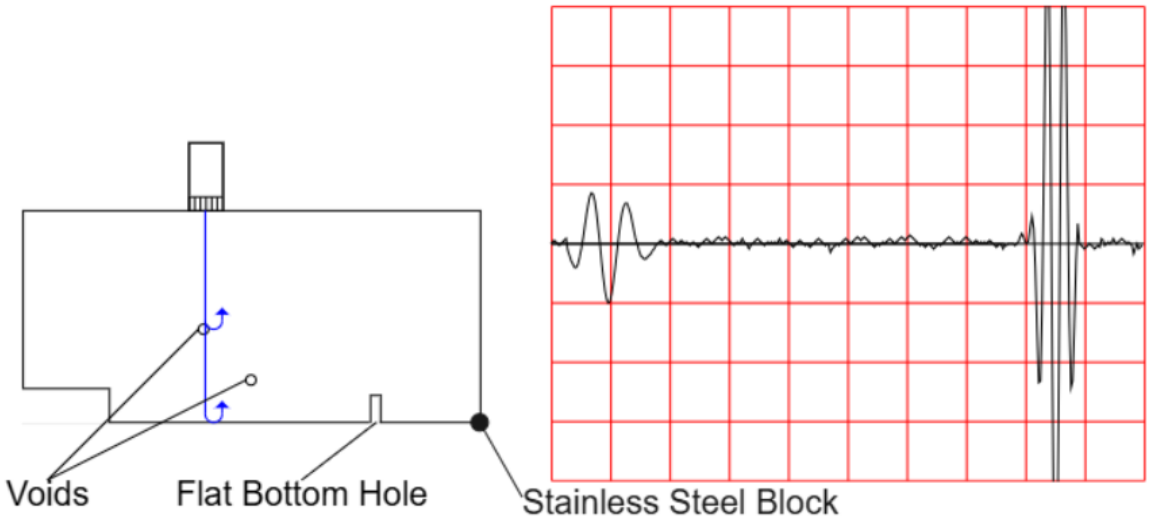


Figure 1 Ultrasonic test basic visual description. The pulse fired by the transducer is partially reflected by one of the voids causing a minor pulse to be observed prior to the main reflection as seen on the right

Phased Array Ultrasonic Testing (PAUT) is a method for testing of larger, more complex structures. This technique utilises a series of piezoelectric elements which can fire at independent time intervals, constructing a resultant wave whose shape and angle of incidence can be altered - Figure 2 [9]. Using PAUT allows comprehensive data acquisition and image construction, ensuring proper defect characterisation while reducing the overall inspection time.

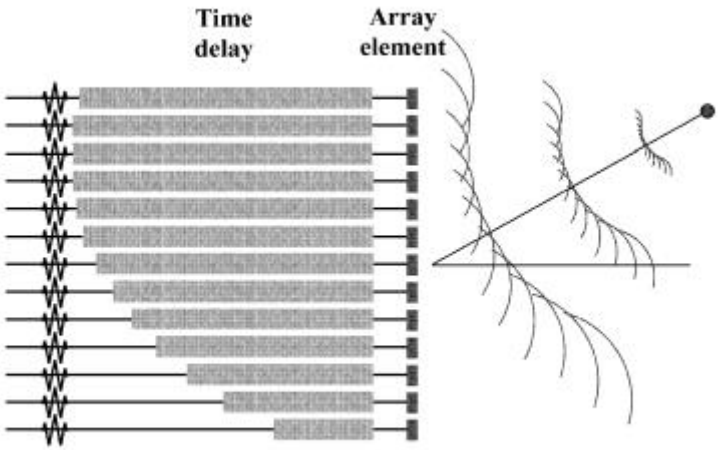


Figure 2 Phased array ultrasonic wave construction. On the left, the different probes with their individual time delays can be seen, with the overall resultant waves and their direction shown on the right

1.1.2 NDT systems

Automated NDT systems are at their most beneficial when operating in hazardous environments, such as at extreme heights, underwater, or over large distances which would take an excessive amount of time for manual inspection. The inspection environment directly effects the type of locomotion and testing technology which is utilised, some examples of general categories are outlined in the following section.

Ground-based robots

These robots maintain contact with the ground and possess a wide range of forms of locomotion, such as snake like crawling, rolling on wheels and walking. [10] Modern technological advancements allow these robots to climb and navigate walls using suction caps and magnetic wheels. An example of such a system is shown in figure 3 [11], a semi-automated inspection device with magnetic adhesion and tracked locomotion:



Figure 3 Tracked wall-climbing robot for ship inspection. Possesses magnetic tracks as shown and is equipped with an ultrasonic probe for NDT. The device is also compatible with other NDT instruments

Airborne robots

Unmanned Ariel Vehicles/Systems (UAVs/UASs) are a recent development within the automated NDT industry. These robots have particular utility in pipeline inspection, being

able to perform large chunks of data collection on long stretches of pipe over short time periods. They can utilise either fixed wings or a rotary blade depending on the application. Figure 4 [12] shows a diagram of the Volio Tricopter, an example of a rotary blade system. A challenge for airborne robots is maintaining stability while undergoing contact tests. The tricopter's tiltable blades allow it to maintain stability while experiencing significant external forces.



Figure 4 Volio Tricopter diagram. This diagram visualises the robot's tiltable blades and includes a surface which is making contact with the robot end effector

Underwater robots

With commercial divers being exposed to many risks, such as drowning, respiratory and circulatory risks, hypothermia, low visibility, and physical injury [13], adequate technology is essential for underwater NDT inspections.

Underwater Unmanned Vehicles (UUVs) are capable of inspecting components in underwater locations which pose the above risks to humans. UUVs are becoming increasingly important due to constantly developing underwater infrastructure. An example of one of these systems is shown in figure 5 [14]:

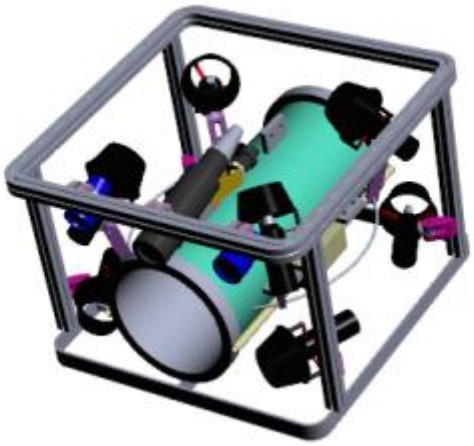


Figure 5 UUV prototype. Possesses a neutrally buoyant rigid body and a series of phased array probes for conducting NDT

1.2 Review of related work

Path planning within robotics refers to control of a robot end effector to follow a desired trajectory or navigation of an entire robotic system [15]. Traditionally, this is done using a predetermined path defined by a series of via points which the robot or end effector must pass through. Motion through via points must be blended to avoid instantaneous changes in velocity, most robots are pre-programmed with this feature.

Navigating a robot within a populated environment requires knowledge of the robot's surroundings. Algorithms such as Simultaneous Localisation and Mapping (SLAM) [16] allow robots to learn their surroundings using real-time sensor data and process history. Various algorithms then exist for optimal trajectory planning, such as A-Star, Dijkstra, D-Star Lite, RRT, neural network, intelligence algorithm, and artificial potential field method [17].

Complex geometries are predominantly inspected using manual methods due to a lack of flexible/dexterous commercial robotic solutions in the current market. Further research is required to integrate complicated mathematical models and detailed sensor data into robotic NDT systems. One of the aims of this project is to utilise the Meca500 [18] robotic arm to perform an accurate scan of a saddle weld, an example of which is shown in figure 6:

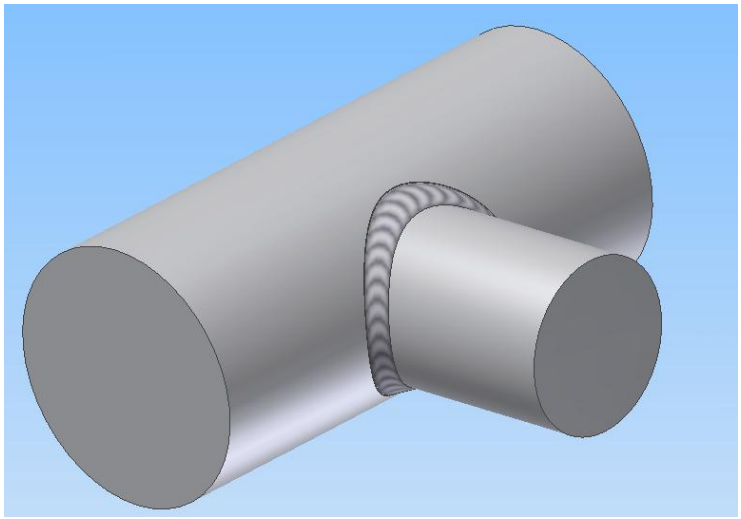


Figure 6 Pipe saddle weld 3D model

This will include the use of an algorithm to move the end effector along the required path, with feedback from a force-torque sensor ensuring consistent and stable contact with the pipe surface. The complete system will include mathematical elements, as well as force-torque and point cloud (or visual) feedback. The following section outlines various methods which utilise these techniques for robotic path planning.

1.2.1 Software based trajectory planning

User defined guideline algorithm

Many robotic systems implement path planning algorithms largely defined within software, i.e. do not rely on extensive sensor feedback, but rather digital twins such as CAD models, or some user-defined parameters. These are advantageous and can produce very accurate scans when the environment is complex, but unchanging and largely understood. Single-sensor single-algorithm systems are severely limited in their ability to resolve ambiguities, identify spurious information and to detect errors or failure meaning that additional forms of feedback are highly beneficial [19].

The presented system from the Hong Kong University of Science and Technology [20] takes advantage of the kinematic parameters of a 2 degree-of-freedom (DOF) stylus (i.e. low moment of inertia) to create an efficient sweep scan based on a guideline specified by a user. An outline of the method is as follows:

1. User specifies guideline and length of stylus.
2. Algorithm calculates head trajectory.
3. Algorithm determines stylus intersection curves with surface for each point of head trajectory.
4. Algorithm determines tip trajectory using sinusoidal formula.
5. Algorithm checks path viability and iteratively applies modifications until complete viability is achieved.
6. Path is output.

The method proved to be more efficient than a zig-zag scanning path, or the iso-parametric method described in [21].

In spite of its advantages, issues arise when considering the amount of user involvement required. For example, surfaces must be manually split into segments before the algorithm is

applied to ensure they can be scanned continuously, and the efficiency of scans depends highly on the provided guideline. Automatically generating guidelines could provide potential improvements. One method to achieve this could use point cloud data of a scene and apply filtering techniques to search for scannable surface regions. The overall accuracy would be improved with additional forms of feedback, such as force and torque or real time visual feedback.

Flood fill and Dijkstra optimisation algorithm

The method proposed from Brunel University London [22] provides a novel approach to path planning, combining the flood-fill and Dijkstra optimisation algorithms to produce autonomous raster scans of surfaces.

Typically scanning surfaces can be defined by a set of nodes and accessible/inaccessible regions. The proposed algorithm takes a deconstructed CAD model containing node information as an input and calculates an optimal raster scan path. A path is created on a node-to-node basis, considering the distance between adjacent nodes and the current node and the intended direction of travel from the current node. A weighting is applied to every node based on these parameters to decide the subsequent node in the trajectory. If the robot reaches a dead end, Dijkstra's algorithm takes over to determine the optimal path to the nearest unchecked node.

This algorithm allows for autonomous sweep scans of complex surfaces. The main factor of importance within this project is the positioning of the ultrasonic probe, which could be optimised by deconstructing the saddle weld component into a point cloud and determining the optimal end effector orientation for each point. Ways to optimise continuity have also been explored.

1.2.2 Visual based trajectory planning

For environments which are unknown and must be learned by a robot, a visual approach is a reasonable choice. Visual based path planning methods can range from being entirely based on learning an unknown environment, to providing additional feedback to improve upon partially pre-defined paths. The reliability of visual feedback is a significant area of research, with visual techniques relying heavily on adequate lighting conditions [23].Some implementations of visual based algorithms are outlined here.

Basic visual robotic navigation system

Many robotic systems operate based on visual feedback, such as that proposed in [24] from the Maharaja Institute of Technology, which presents a simplistic visual based approach to navigating a robot within a populated environment. The system is able to create an image of the floor by combining a series of frames, then using the laser can detect obstacles and translate this information into basic avoidance instructions. Scenes are stored in a text file to create a map of the environment.

The system exhibits a number of limitations, such as a lack of sensitivity to uneven surfaces, largely approximated motion for obstacle avoidance and overall lack of versatility. This highlights the need for additional forms of feedback and more complex/precise path planning algorithms.

Point cloud camera integration for reliable ultrasonic inspection

In order to meet industry standards, it is essential that NDT methods are reliable and not compromised by innovation [25]. In the case of ultrasonic testing, reliable inspections require

the probe to maintain consistent contact with the test surface. In order to properly position the probe, it is necessary to have good knowledge of the test surface. Surface feature detection can be divided into contact and non-contact methods, with many current methods being non-contact [26], such as visual feature detection using a camera.

The method proposed in [27] from Inner Mongolia University of Science and Technology uses a point cloud camera integrated onto a robot to create 3D representations of components. A number of steps are outlined, such as hand-eye calibration to establish the robot's knowledge of the camera pose relative to its end effector, image pre-processing upon acquisition of a signal, including various filtering techniques such as pass through, radius and statistical filtering, followed by determination of surface normals using the local plane fitting method. Having obtained a digital representation of a physical component which includes surface normal vectors, accurate ultrasonic scans can be conducted by maintaining the approach vector of the end effector parallel to the surface normal at each inspection point. The article then verifies the accuracy of the system using a mathematical model comparison.

1.2.3 Force-torque sensor integration

With the orientation of the probe for PAUT being of high importance, various methods exist to improve the accuracy of probe alignment, such as the use of a laser scanner or ultrasonic emitter [28]. A method which will be explored in this project is the integration of a force-torque sensor onto the payload to provide tactile feedback, ensuring that the probe maintains consistent contact with the surface.

An attempt at this concept was undergone at London Sount Bank University [29]. The end effector, including the force-torque sensor, of the designed robot for this application is shown in figure 7 [30]:

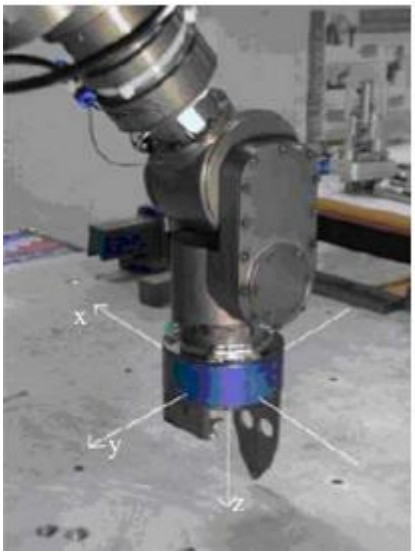


Figure 7 NDE system with force-torque sensor for additional feedback

This control method involves maintaining the contact force at a particular threshold value and minimising the torque.

The article highlighted the benefits of fully automated NDT. Manual inspections on complex/uneven surfaces were compared to inspections conducted by the robot, with the automated inspections performing better, particularly at low speeds (~0.5 mm/s).

In spite of these extremely low speeds, it is clear that this control method has potential. Further optimisation would aim to increase these speeds to a point where automated tests are unequivocally more efficient than manual tests.

1.2.4 Trajectory planning using ultrasonic feedback

Ultrasonic testing is widely applied to examine the following characteristics of welds [31]:

1. Weld integrity
2. Weld structure
3. Weld strength
4. Weld elasticity/shearing strength

Due to many welds lacking visibility [32], visual identification of welds can lack reliability. Ultrasonic data provides a more detailed characterisation of a material, describing its internal state. Therefore, such data can be used as a source of feedback for more accurate path planning.

Brunel University London presents a system [33] which uses ultrasonic data as a means of weld tracking. This system uses PAUT to create a series of images as the probe moves along a surface. In order to identify and track welds, the system uses the mean shift algorithm [34] to map characteristics of the previous frame to the current frame. The system was proven to work with reasonable accuracy, highlighting the advantages of this method for path planning over the use of visual based feedback. Only two very simplistic weld configurations were tested, meaning the method is yet to be confirmed suitable for complex geometry. The possibility of using the mean shift algorithm on ultrasonic images to track a saddle weld will be considered. This would require adapting this algorithm to operate on a 3D surface rather than 2D.

1.2.5 Saddle path trajectory planning

With of the aims of this internship being a successful scan of a saddle weld, algorithms for planning this exact motion path are of high relevance. The following system proposed by a group of researchers from various institutions in China [35] consists of a 3D laser-based reconstruction algorithm to generate a trajectory path for a multi-layer saddle weld.

Following point cloud generation, the method applies a RANSAC [36] cylinder fit, establishing the location and orientation of the two conjoined pipes. The intersection of the axes of the two pipes is the location of a user co-ordinate system which all path points are described with respect to.

The weld cross section varies with respect to its angular position about the smaller pipe as shown in figure 8:

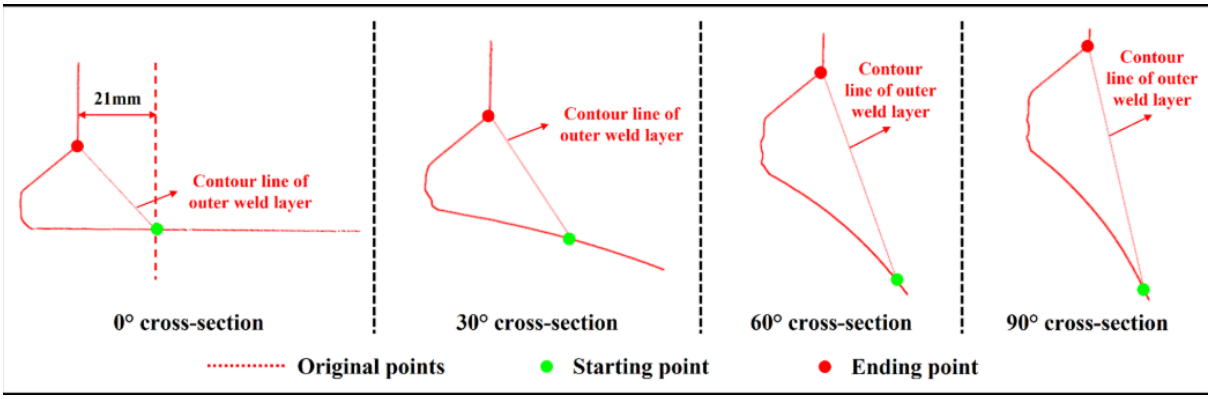


Figure 8 Variation in cross section of saddle weld with respect to angular position about smaller pipe.

The system divides the weld into 24 cross sections, further dividing the cross sections into layers then pass points. Each cross section contains the same pass points so that paths can be generated by matching corresponding pass points.

This method was shown to produce accuracies within 1 mm. This validates the use of RANSAC cylinder fitting as a possible method for generating a saddle curve from a point cloud. The user co-ordinate system at the intersection of the pipe axes provides a universal reference, greatly simplifying the path planning process. This idea was applied in this project.

A potential addition to the system is a method for calculating path feasibility based on the work envelope of the robotic arm.

1.3 Review of NDT systems at Strathclyde

This section discusses some projects at the University of Strathclyde which are of relevance to this internship and encompass some of the path planning techniques highlighted in the previous section.

1.3.1 System 1 – enhancing positional awareness

A significant issue within automated NDE is positional awareness [37]. In order for industry standards to be sufficiently met, and for NDE tests to be carried out efficiently and reliably, a robot's knowledge of its position is crucial.

The following system, designed with the University of Strathclyde, is an application which utilises the Meca500 mounted upon a wheeled platform to create a ground-based inspection robot, shown in figure 9 [38]:

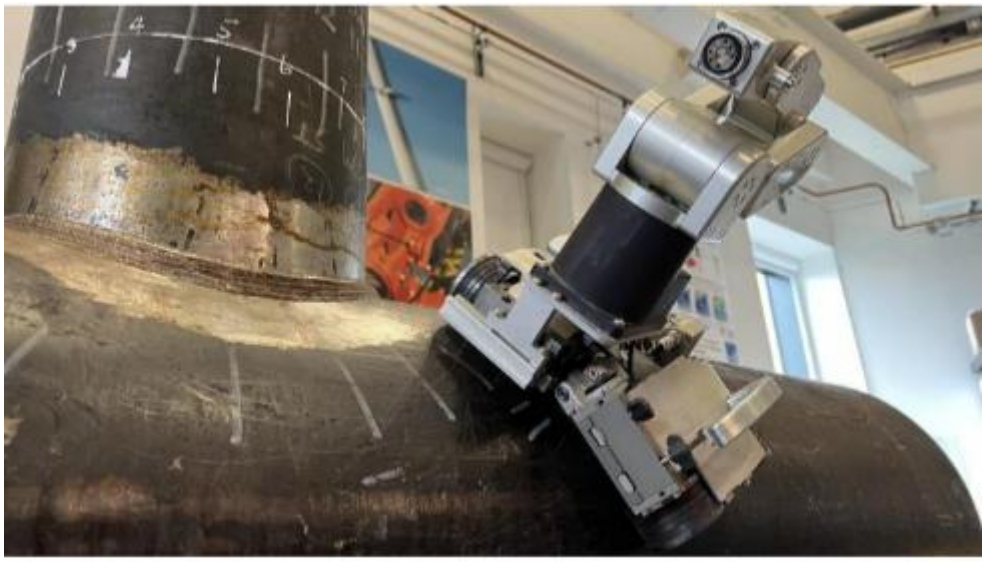


Figure 9 Meca500 arm mobile NDT system application. As can be seen the robot can maintain adhesion on a pipe surface

This system addresses several issues regarding autonomous NDE, such as complicated setup procedures, inspection time and localisation. The magnetic platform allows the robot to access difficult to reach locations, with the 6 DOF Meca arm providing the necessary means to undergo NDE inspections. The device is equipped with four cameras, achieving a field of view which covers 360 degrees providing the robot ample coverage of its environment.

A saddle weld inspection algorithm was implemented into this system, similar to that which will be explored within this project.

Improvements to this system include the utilisation of a point cloud camera to construct a 2D image of the area just in front of the platform upon which a user can select a series of points to define a desired scan. This would further enhance the versatility of the robot as well as its positional awareness.

In order to further improve the accuracy of scans, the article suggests the installation of a force torque sensor onto the end effector to provide feedback regarding contact between the payload and the surface.

The system highlighted in this paper provides the groundwork for this body of work, as it is an aim to utilise this robotic arm (Meca 500) to conduct a scan of the saddle weld sample shown in figure 6.

1.3.2 System 2 – combination of RGB camera data and force-torque sensor feedback

This system is an advancement of system 1 and provides further groundwork for this project.

As highlighted previously, maintaining consistent perpendicular contact between a probe and a test surface is essential for accurate ultrasonic tests. System [27] proposed a method to accurately determine surface normals of a constructed point cloud image. System [30] utilised a force-torque sensor to apply rotational corrections to the end effector when in contact with a surface. The following system at the University of Strathclyde [23] integrates these two forms of feedback together to further mitigate errors.

A lightweight point-cloud camera is used in order to minimise the weight of the payload. The compactness and rapid data capture of such cameras is highly advantageous, however variable lighting conditions lead to inaccuracies in the constructed images. The potential of using a laser scanner as an alternative to a visual point cloud camera will be explored to mitigate the effect of poor lighting and inaccurate surface recognition. With ultrasonic scans requiring precise probe alignment, the addition of the force-torque sensor provides the necessary feedback to correct the orientation to within the allowed range by means of a PID controller.

1.3.3 System 3 – improving probe positioning using ultrasonic imaging

In relation to the issues raised in [32] and [33], an ongoing challenge is ensuring accurate detection and tracking of welds which are not visible to the naked eye. The following application, also from the University of Strathclyde, utilises PAUT to create images describing the internal characteristics of components which are then used to accurately position the probe by recognising exactly where a weld is situated. Figure 10 [39] shows an example of a constructed image of a weld cross section from a series of ultrasonic tests:

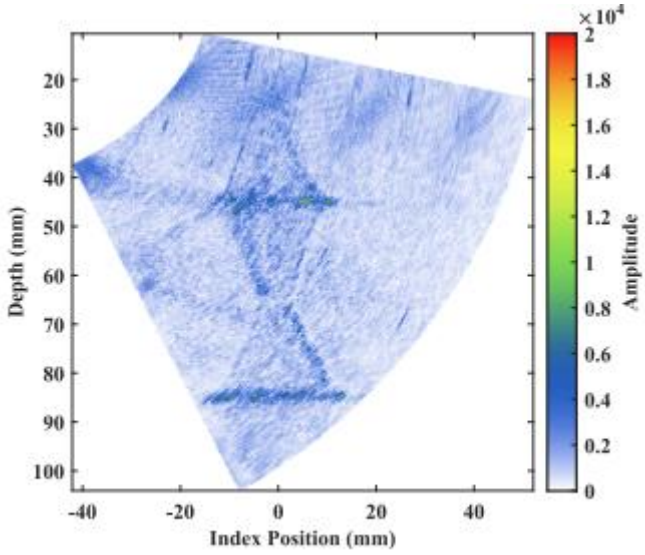


Figure 10 Constructed image of weld internal cross section using ultrasonic tests. The weld is invisible to the naked eye but its shape is clearly seen in the figure

This system is a good example of the depth of information than can be attained using ultrasound, highlighting the benefits of using this method for NDT.

Once the capabilities of this system have been enhanced to account for a wider range of weld types and materials, it could be implemented onto a similar locomotion system to that of [38].

1.3.4 System 4 – multi-sensor robotic inspection system

The trajectory planning techniques outlined in the previous section are all advantageous for specific applications, however it is in the combination of software defined techniques and multi-sensor feedback that robustness and reliability can be maximised [19]. This project aims to achieve this through combining point cloud and force-torque data with software defined path generation and control algorithms.

Another multi-sensor path planning method is proposed by the university [40]. Considering the benefits of multi-sensor feedback, this system combines RGB-D point cloud data with

feedback from a force sensor and laser scanner to conduct scans on largely curved surfaces.

With visual based point cloud data lacking precision and versatility due to its sensitivity to lighting conditions and varying surface types [41], laser scanning is incorporated to maintain correct normal orientation of an ultrasonic probe with a surface following scan path generation. The system uses feedback from three laser scanners to apply real-time corrections to the probe orientation when following a generated path from erroneous point cloud data.

Figure 11 displays generated waypoints for a raster scan based on RGB point cloud data:

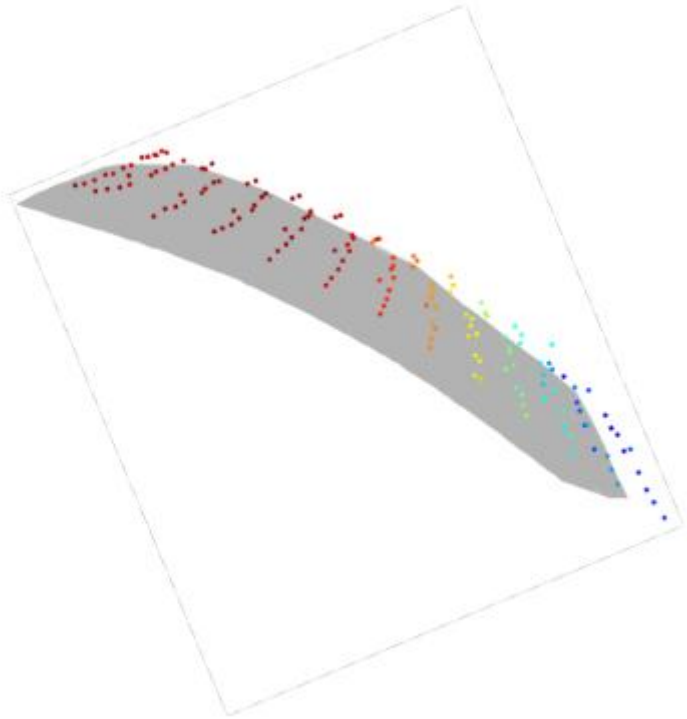


Figure 11 Generated waypoints for raster scan on curved surface [40]

As can be seen, the waypoints exhibit significant deviation from the surface.

In addition to this, force control is implemented to monitor contact pressure, with this being of significant importance for ultrasonic NDT [42]. Overall, the system comprised a P controller for tangential velocity and force control, and a PI controller for normal force control so as to maintain a global constant value.

A saddle weld comprises a more precise scanning path than a free form surface, meaning that waypoint generation must be more robust. Orientation control using torque rather than laser feedback is explored within this body of work, as well as the possibility of using laser scanning to generate the entire point cloud.

2 Scope of work

The goal of this body of work was to develop suitable control algorithms to allow manipulator based remote inspection systems, such as the example shown in Figure 12, to inspect complex geometries such as saddle welds, Figure 6.

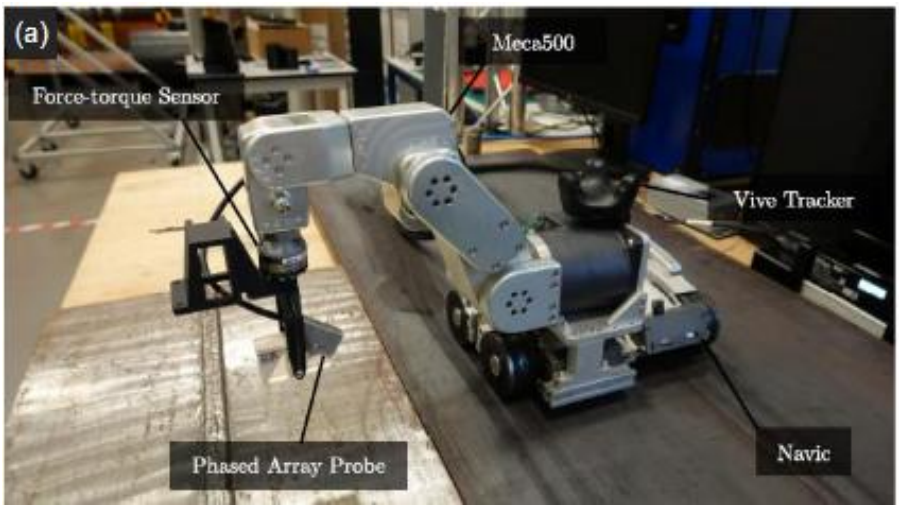


Figure 12 Remote inspection system where the base carries a small robotic manipulator with an ultrasonic payload

The primary goal of controlling a robotic arm to follow a complex inspection path along the surface of a saddle weld provides a basis for future development where the base will move to various positions around the weld. The arm will then be responsible for following the path defined within inspection standards. For this body of work, research will focus on the control of the manipulator only, providing the research group a greater level of understanding of the difficulty of the task and its nuances.

A summary of the project output is described below:

- 3D printed pipe section for mock inspection
- Path generation algorithms in MATLAB for scanning above and below weld
- PID control-based algorithm for path trajectory execution
- Force-Torque sensor integration to ensure proper coupling and proper alignment is maintained throughout the scan
- 3D printed mount for ultrasonic probe and laser scanner
- Point cloud generation algorithm of real pipe sample using laser scanner
- Adapted path generation algorithm based on point cloud data

3 System overview

3.1 Physical setup

This body of work was carried out using the MECA500 robotic arm (figure 13) [43] from Mecademic Industrial Robotics [44].



Figure 13 Meca500 robotic arm and power box used to carry out work

The arm was attached to a metal board and a model of a pipe intersection between a 160 mm radius and 250 mm pipe was designed in Solidworks and 3D printed for mock inspections. Figure 14 (a) shows the CAD model of the mock inspection unit and figure 14 (b) shows a real-life image. 3D drawings of the mock pipe section are contained within appendix A.

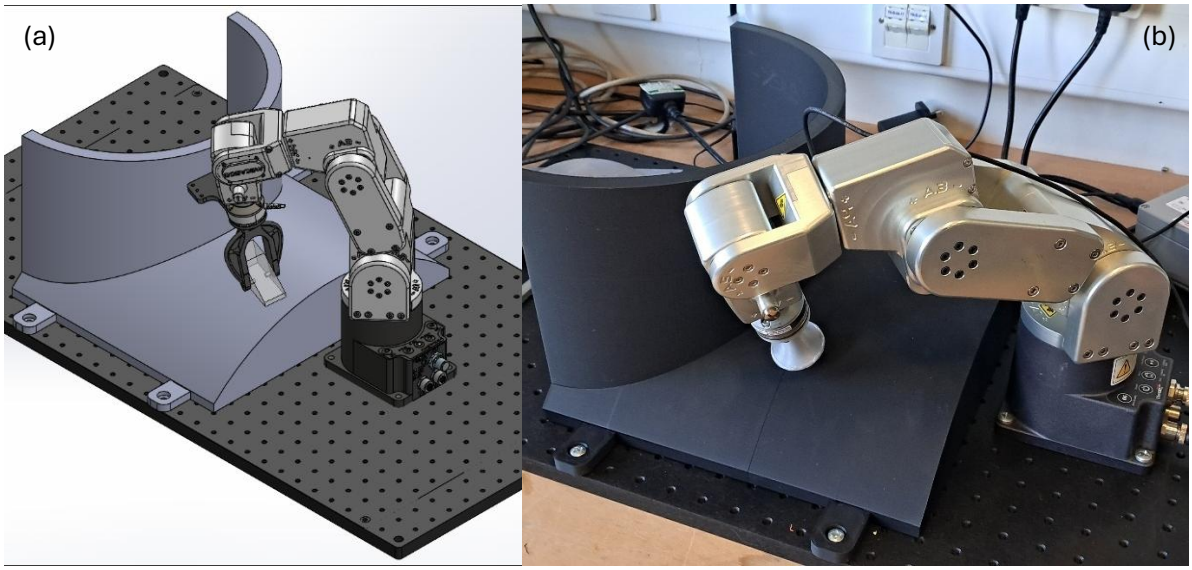


Figure 14 (a) CAD model of mock inspection unit with pipe section and Meca500 arm equipped with ultrasonic probe wedge and FT sensor (b) real-life setup with dummy probe equipped

Figure 15 displays the setup integrated onto an actual pipe using the NAVIC magnetic roller [45].

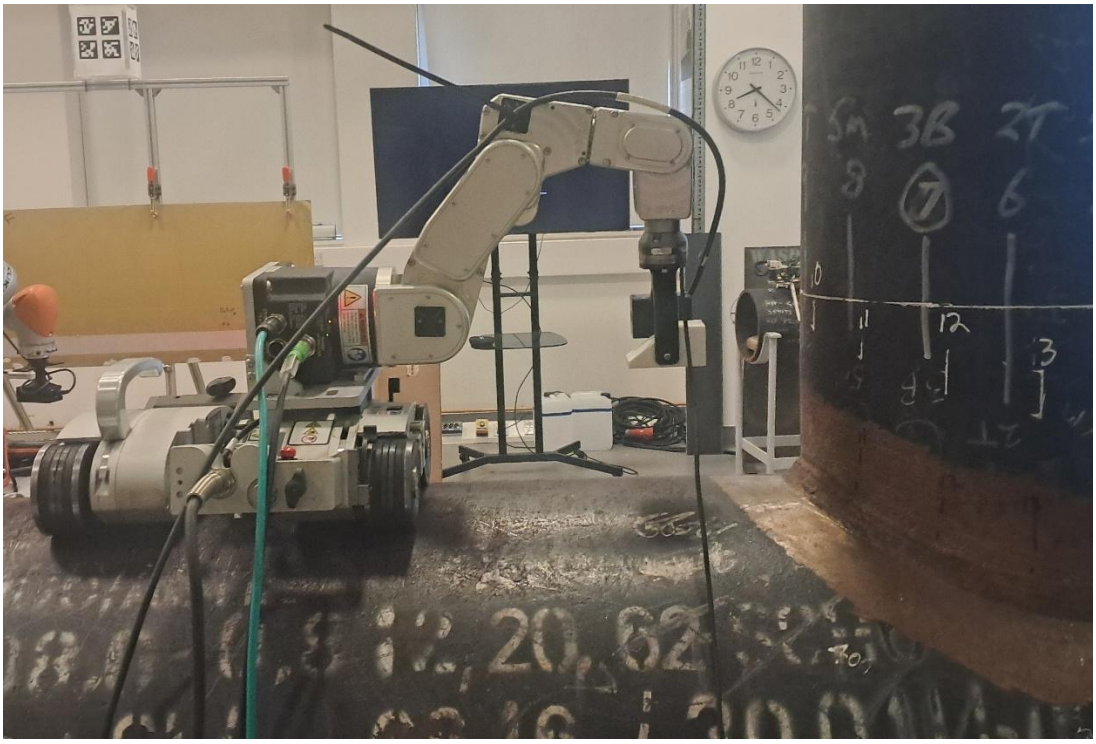


Figure 15 System set up on real life pipe sample using magnetic roller probe

The ATI multi-axis force-torque sensor [46] and Micro-Epsilon Opto1420 laser [47] were used for the system. An appropriate mount was designed to combine these instruments with a phased array ultrasonic probe and wedge by Eddyfi Electronics [48] onto the robot end effector, shown in figure 16.

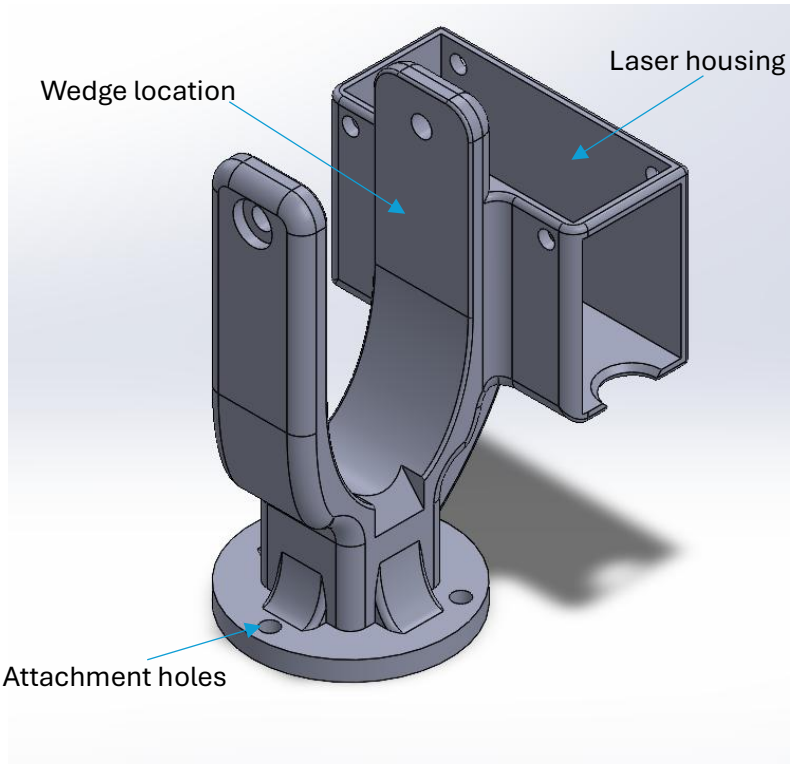


Figure 16 CAD model of mount for phased array probe and Opto1420 laser

A 3D drawing of the mount is included in appendix B.

The robotic arm was programmed on VScode using the Python language and the provided package for the Meca500 [49].

3.2 System operation

The inspection path requires a consistent 40 mm offset from the front of the ultrasonic probe and the weld. The robot tool reference frame (TRF) was positioned on the front centre of the probe and its orientation aligned with the x and y axes of the force-torque sensor. This required the following translations from its default position: 31.9 mm along the x-axis, 113 mm along the z-axis and 90 degrees about the z-axis.

The system is summarised by the steps shown in figure 17. Each of these steps is described in this section.

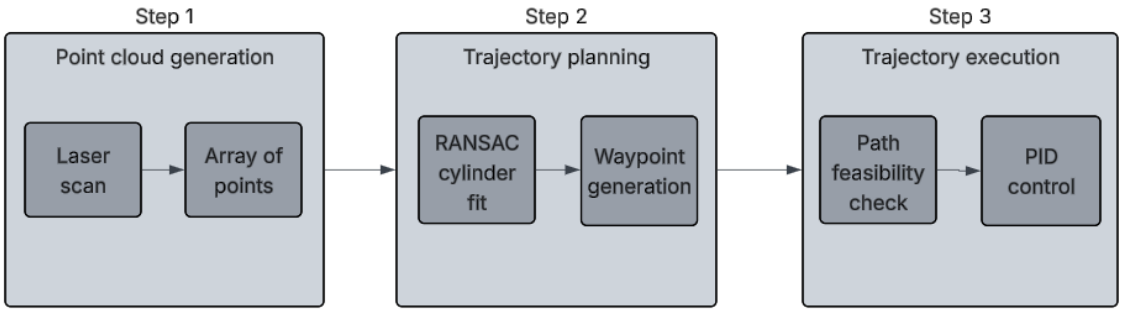


Figure 17 Process summary

3.2.1 Point cloud generation

The location of the laser point on the Opto1420 laser was obtained from the data sheet and this was used to determine its offset from the TRF (figure 18).

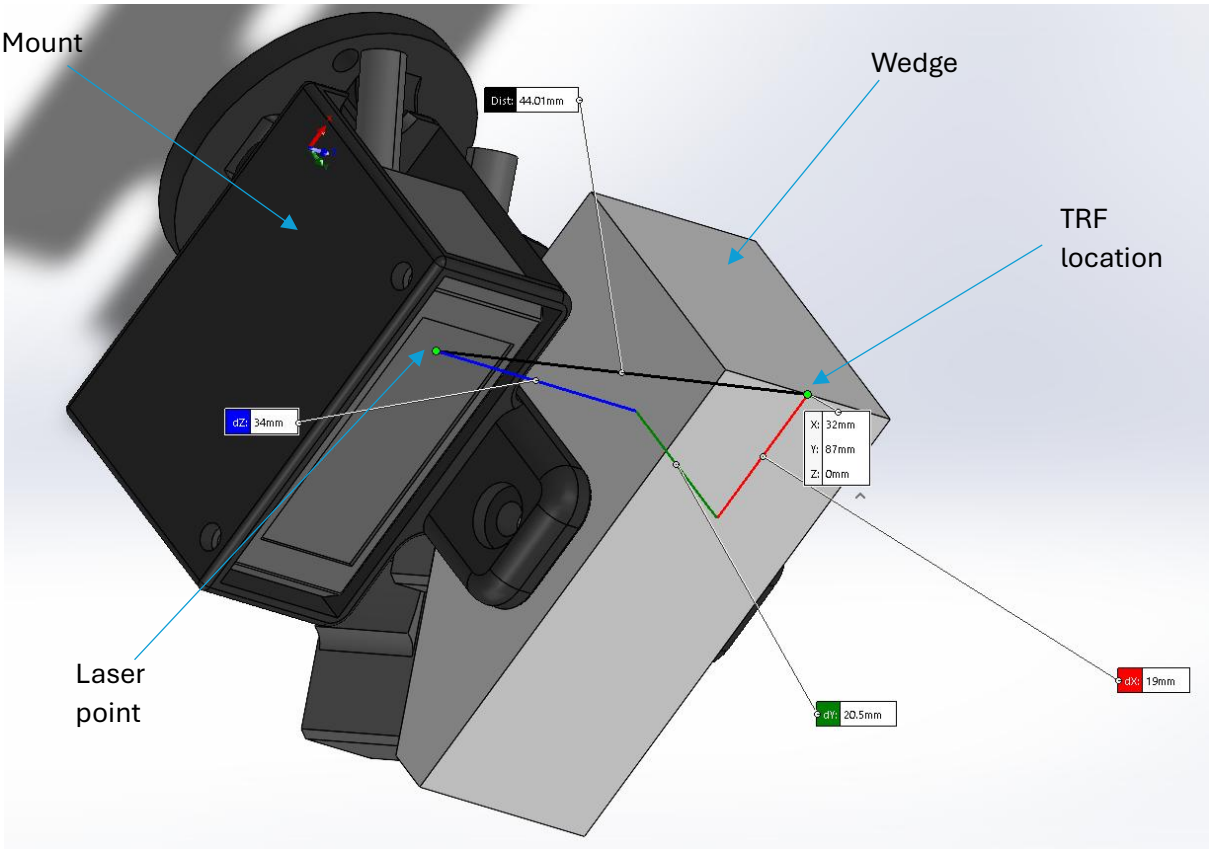


Figure 18 Mount with laser and wedge attached showing laser point offset from TRF. Note that dX, dY and dZ do not correspond to X, Y and Z offsets due to the TRF having a different orientation

The DLL file for interfacing with the Opto 1420 laser was downloaded from the Micro-Epsilon website [50] and the DSensor_ILD1420 class for calling the DLL file from a Python script was adapted from the following website [51]. The following line of code was used to reference the path to the DLL file:

```
windll.LoadLibrary(r'C:\Users\TIC714 Guest\Documents\MEDAQLib-5.3.1.392\Release-x64\MEDAQLib.dll')
```

The remainder of the source remained unchanged. The laser distance measurement is accessed through the 'poll' function.

The poll function within the class provides the laser distance measurement with an offset of 100 mm (the lower limit of the working range of the laser) so 100 mm was added to all measurements.

In order to generate a point in 3D space from a laser distance measurement, kinematic relations are used. Since the laser aligns with the tool frame z-axis:

Coordinates of a point in TRF = (x offset, y offset, z offset + measured distance + 100 mm)

Equation (1) was used to transform points within the TRF to the robot base frame or world reference frame (WRF).

$$^{A}P = {}^{A}_{B}R.\,{}^{B}P + {}^{A}P_{Borig} \qquad (1)$$

So, if a point within the robot TRF ^{B}P is obtained, it can be transformed into a point within the robot base frame ^{A}P using the rotation matrix $^{A}_{B}R$ and translation vector $^{A}P_{Borig}$ between the

robot base and tool frame at that instant in time. These can be obtained using the built-in kinematics of the Meca500. Note that the software describes the orientation of the tool frame in terms of Euler angles so the transforms3d [52] library was used to convert these to a rotation matrix.

Generation of a point cloud involved a data collection process of laser distance measurements to the pipe section, applying equation (1) to each measurement to generate the coordinates of each point within the point cloud with respect to the WRF. To achieve good coverage of the pipe section, the robot was positioned on top of the horizontal pipe and conducted four separate scans corresponding to the left and right side of both pipes. A scan involved motion of two joints, one oscillating between upper and lower limits and the other performing a single sweep. Rotatory joints were typically joints 4 and 5. The four scanning positions were dictated predominantly by joint 1. Table 2 outlines the joint values and ranges for each of the scans.

Table 2 Joint parameters for laser scans

Key: > = single sweep between values <> = oscillates between values		**Side**			
		Right		**Left**	
Pipe	**Vertical**	**Joint**	**Value (deg)**	**Joint**	**Value (deg)**
		1	0	1	-175
		2	-65	2	-65
		3	-95	3	-95
		4	-20 > 45	4	-35 > 25
		5	90 <> 115	5	90 <> 115
		6	90	6	180
	Horizontal	1	-45	1	-135
		2	-70	2	-70
		3	-40	3	-40
		4	-65<>-35	4	5<>45
		5	70>115	5	70>113
		6	-90	6	-90

Note that these parameters were optimised for the robot's position with respect to the pipes (on top of horizontal pipe with approx. 500 mm offset from vertical pipe). The parameters should be altered depending on where the robot is situated.

The Pyvista [53] library was used to visualise the point cloud once generated, as shown in figure 19.

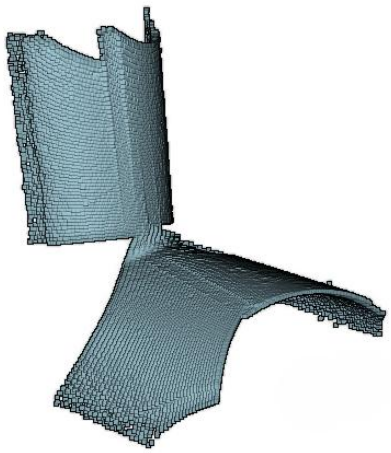


Figure 19 Example of generated point cloud from laser scan displayed using PyVista library

Different scanning techniques are discussed in the testing section. The function within the program for the point cloud generation process is called swing_scan_circumference and comes under the ForceControlScanning class along with all other robot control functions.

3.2.2 Path generation

Having obtained a point cloud image of the pipe section, the program is required to conduct ultrasonic scans from both sides of the weld. Figure 20 shows scan paths 1 and 2.

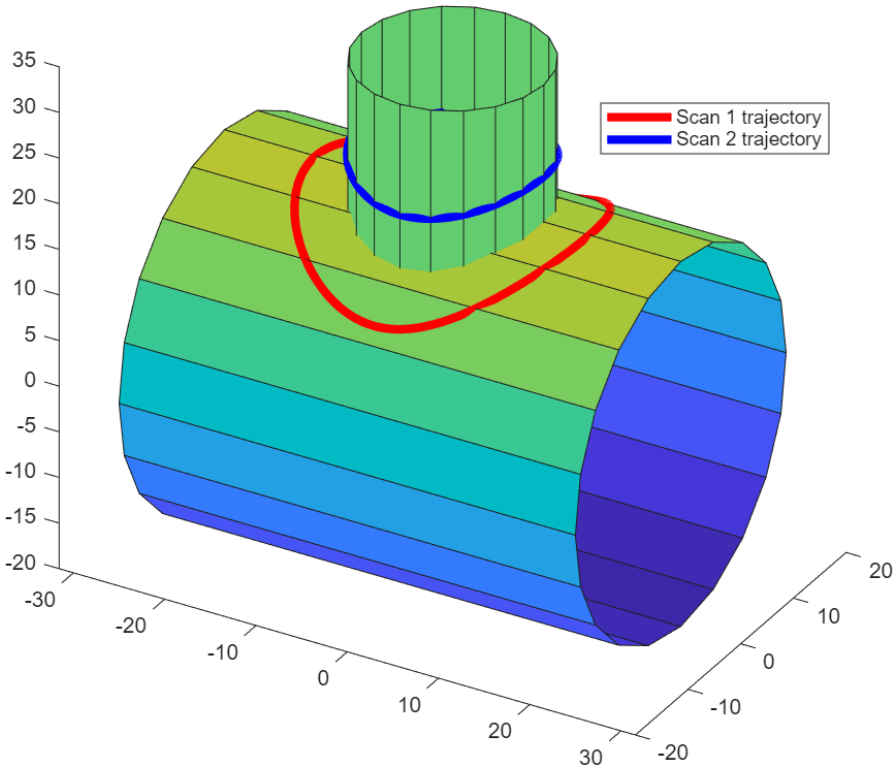


Figure 20 Scan paths 1 and 2 visualised on pipe intersection

In order to generate the path for the end effector to follow, equations were established to describe the TRF location and orientation corresponding to each point in the scan, and the scan was divided into a series of intermediate waypoints. These equations provided a basis for the scan path which would be subject to corrections based on force-torque feedback. The position equations for both scan trajectories took the form of the Steinmitz equations [54] as shown below.

$$X=r\cos(t) \quad (2)$$

$$Y=r\sin(t) \quad (3)$$

$$Z=\sqrt{R^2-y^2} \quad (4)$$

'r' is the radius of the smaller vertical pipe, 'R' is the radius of the larger horizontal pipe and 't' is the angular position about the vertical pipe (0<t<360). These equations describe the TRF position with respect to a coordinate frame situated at the intersection between the central axis of the two pipes as shown by figure 21.

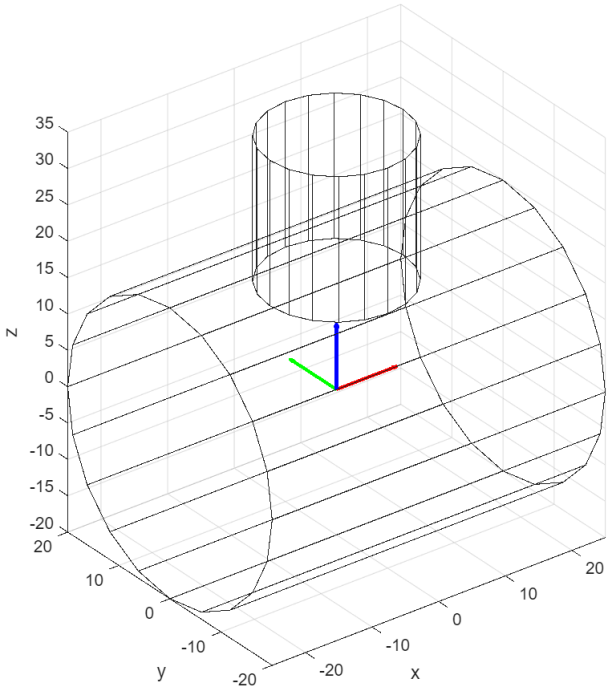


Figure 21 Local coordinate frame at intersection of cylindrical axes visualised

In addition to the requirement of following the paths shown in figure 20, the orientation requirements of the TRF while conducting an ultrasonic scan include:

- Normal vector (z-axis) is normal to contact surface
- Approach vector (y-axis) is normal to weld
- Sliding vector (x-axis) points in the direction of motion

The equations describing the TRF Euler angles took different forms for each scan. For scan 1 (contact on horizontal pipe), the orientation equations were derived in the following way (aided by figure 22)

Waypoint coordinate frames for scan 1 are orientated by rotating the initial end effector position (waypoint 0) about its x axis (roll) so as its z axis sits normal to the horizontal pipe surface, followed by a rotation about its z axis (yaw) so as its y axis is normal to the vertical pipe surface. The initial roll and yaw offsets of waypoint 0 from the local coordinate frame

are added to all subsequent waypoints since the angles have been defined with respect to waypoint 0. These offsets are 180 degrees about the x axis and 90 degrees about the z axis.

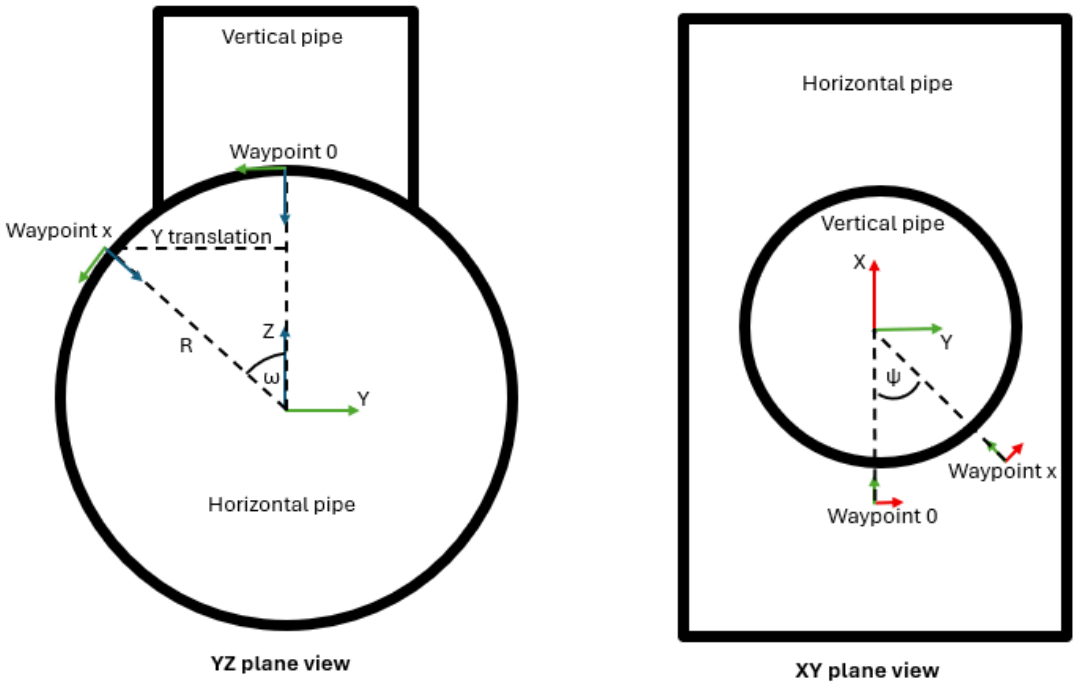


Figure 22 (a) YZ plane view of pipe intersection showing required roll angle for z axis to be anti-normal to the horizontal pipe surface. (b) shows the required yaw angle for y axis to be anti-normal to vertical pipe surface. Roll operation must be prior to yaw

As can be seen, the rotation about the x axis is described by a right-angled triangle comprised of the radius of the horizontal pipe and translation along the y axis of the local coordinate frame, and the subsequent rotation about the z axis is simply the angular position about the pipe. The Euler angle equations describing all waypoints in the trajectory for scan 1 are therefore derived as follows (note that the conventions turn out to be negative):

$$\omega = -\tan^{-1}(y/R) + 180 \quad (5)$$

$$\varphi = 0 \quad (6)$$

$$\psi = -t + 90 \quad (7)$$

For scan 2, determining the Euler angles describing each coordinate frame was slightly more complicated as a combination of rotations about all three axes from the local coordinate frame position are required to establish a waypoint orientation. The process of determining the Euler angles for each waypoint is as follows:

1. Calculate rotation matrix for initial waypoint
2. Apply equations to calculate Euler angles between subsequent waypoints and initial waypoint. Convert these to rotation matrices
3. Combine rotation matrices using matrix multiplication
4. Obtain overall Euler angles from resultant rotation matrix

Figure 23 helps visualise how the Euler angle equations were derived for step 2.

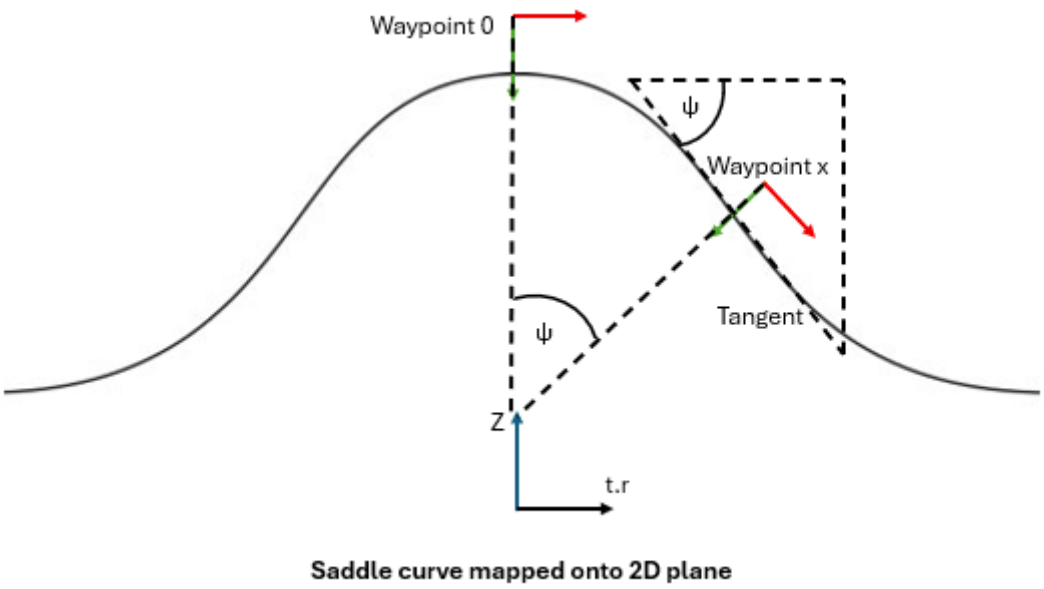


Figure 23 Saddle curved mapped onto 2D plane visualised. This curve describes how the z value changes with respect to position along the circumference of the vertical pipe

As can be seen, the scan trajectory requires that the y axis of the TRF remains normal to the tangent of the saddle curve mapped onto a 2D plane of the z axis of the local coordinate frame against the position on the circumference (angular position 't' multiplied by the vertical pipe radius 'r'). The gradient of the tangent is equal to the first derivative of the z position on the curve with respect to the circumferential position. The inverse tangent function is used to convert this to the yaw angle as shown:

$$\Psi = \tan^{-1}\frac{dz}{dt.r} = \tan^{-1}\frac{d(\sqrt{R^2-y^2})}{dt.r} = \tan^{-1}\frac{d(\sqrt{R^2-r^2\sin^2 t})}{dt.r} = \tan^{-1}\frac{r.\sin t.\cos t}{\sqrt{R^2-r^2\sin^2 t}} \quad (8)$$

The same methodology for deriving the pitch and yaw angles for scan 1 applies to the deriving the roll and pitch angles (respectively) for scan 2:

$$\omega = 0 \quad (9)$$

$$\varphi = -t \quad (10)$$

Applying these equations generated the following TRF paths (figure 24).

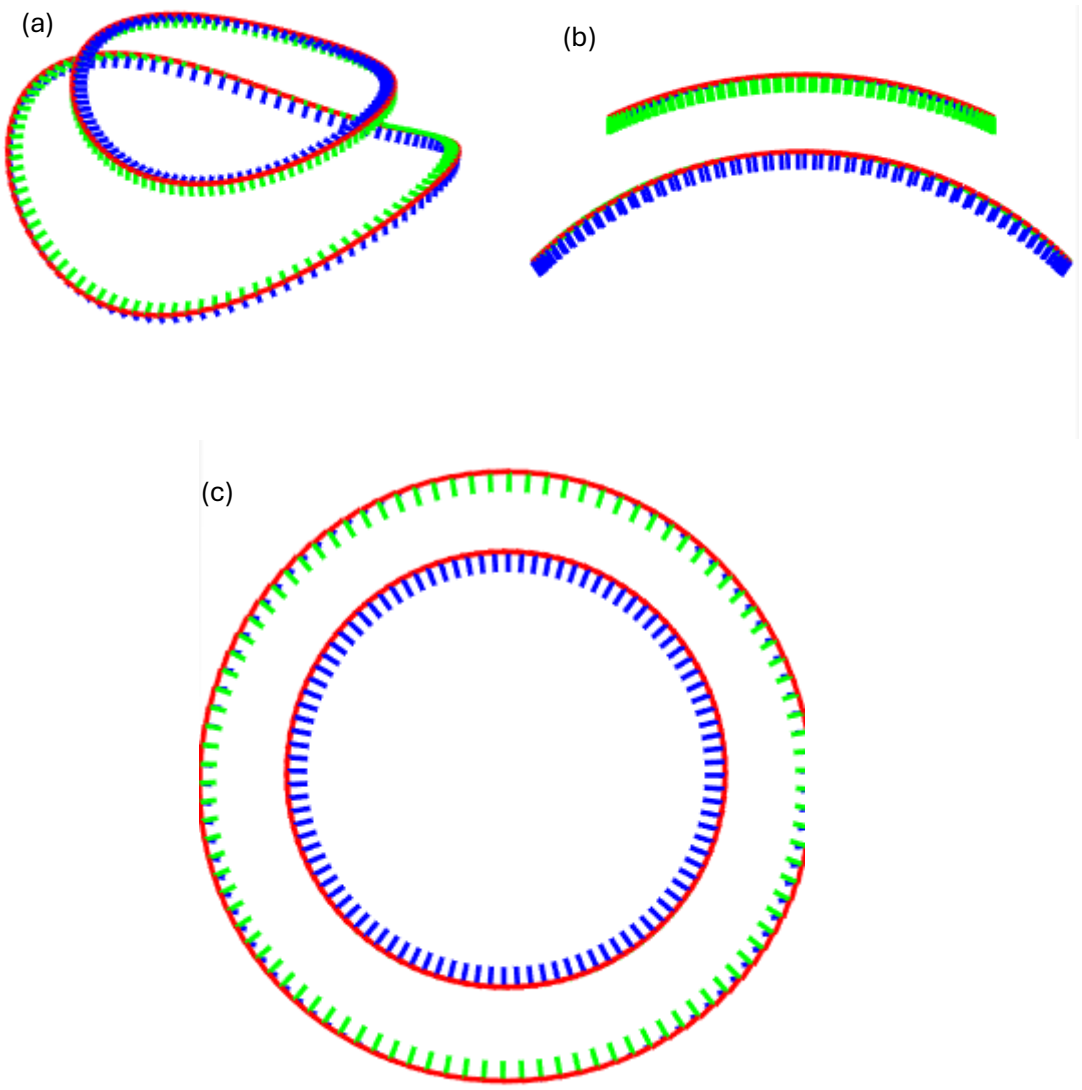


Figure 24 (a) TRF path for saddle inspection 3D view (b) YZ plane view showing tangency to weld and anti-normality to horizontal pipe surface (c) XY plane view showing anti-normality to vertical pipe

In order to obtain values from the equations the radii of both pipes had to be obtained from the point cloud data. This was done in MATLAB using the Computer Vision toolbox [55] to apply a RANSAC cylinder fit to the data set. The MATLAB engine for Python [56] was used to pass data between the two software applications, and the Robotics Systems toolbox [57] was used for Euler angle transformations within MATLAB. To aid the cylinder fit process, horizontal and vertical orientation constraints were applied to both cylinders with some leeway. An example of a RANSAC cylinder fit on some collected point cloud data is shown in figure 25.

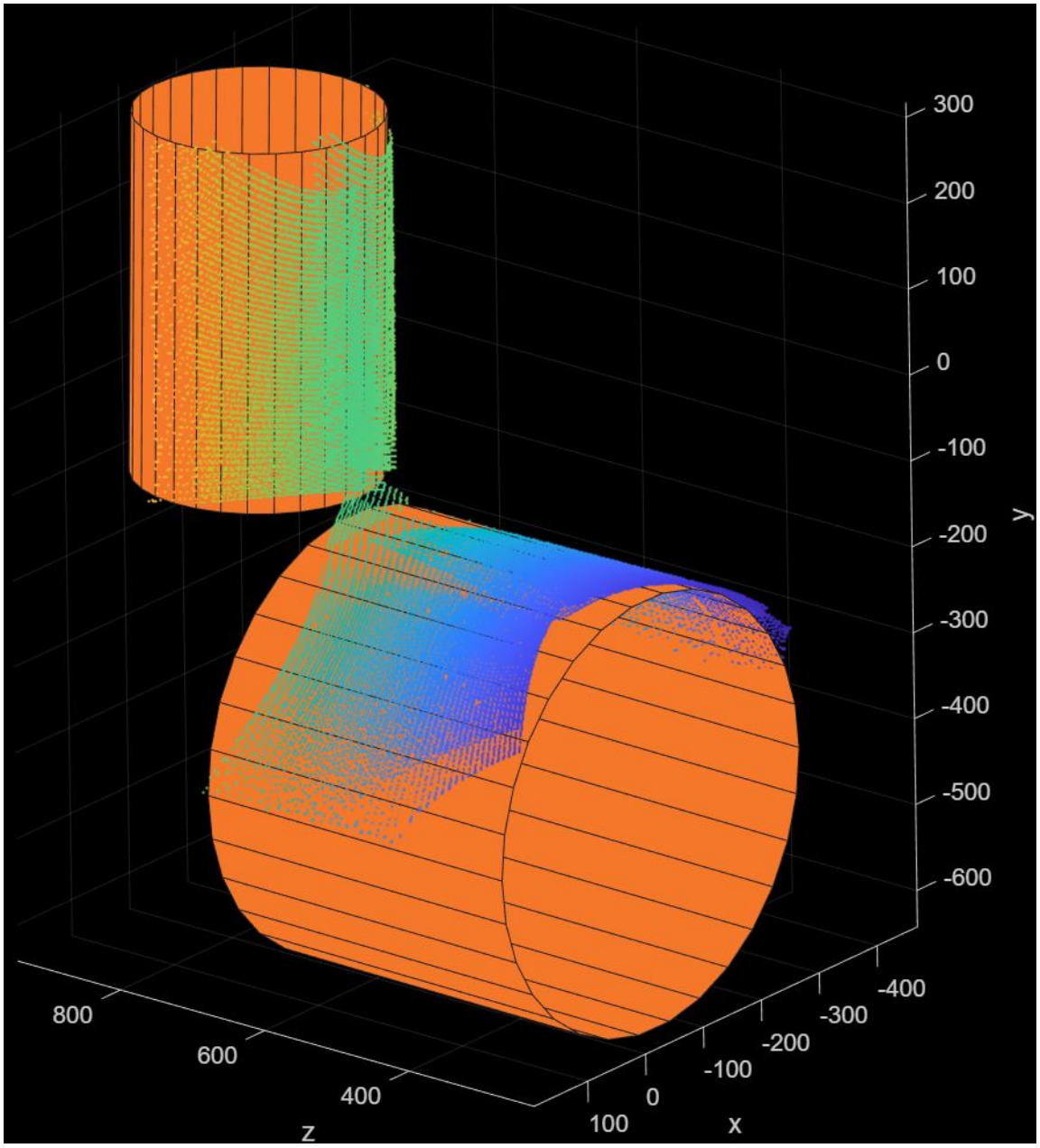


Figure 25 Generated point cloud data from laser scan with RANSAC fitted cylinders

Having obtained data describing the cylinder radii and orientations, the position of the pipe axis intersection could be derived. This is the position of the local coordinate frame. Because it is unlikely that the axes of the fitted cylinders would actually intersect, the nearest points to intersection between the two axes were calculated using the following derivation [58].

Given two lines:

$$L_1=P_1+t_1V_1 \quad (11)$$

$$L_2=P_2+t_2V_2 \quad (12)$$

The requirement is to find two points X_1 on L_1 and X_2 on L_2 such that the distance between X_1 and X_2 is minimal.

The line connecting these points must be perpendicular to both other lines, therefore, let the perpendicular line start from a point $P_1+t_1V_1$ of the first line, and have tangent vector V_3, i.e. $L_3=P_1+t_1V_1+t_3V_3$

It should be that $V_3 \cdot V_2=0$ and $V_3 \cdot V_1=0$, which means that you can get V_3 as $V_3=V_2 \times V_1$

Now for this line to meet also the second line you need to have $P_1+t_1V_1+t_3V_3=P_2+t_2V_2$

With this you have 3 linear equations in 3 variables, t_1, t_2 and t_3

Once you solve for them, then:

- the distance between line 1 and line 2 will be $d = t_3 \cdot V_3$
- the closest points will be $Q_1=P_1+t_1V_1$ on line 1, and $Q_2=P_2+t_2V_2$ on line 2.

Therefore, two coordinate frames located at Q_1 and Q_2 are obtained, the distance (d) between which is equal to the offset between the two cylindrical axes at their nearest point to contact. The orientation of frame 1 is dictated by the axis of the horizontal cylinder (V_1) and the normal vector (V_3), and the orientation of frame 2 is dictated by the axis of the vertical cylinder (V_2) and the normal vector (V_3). Figure 26 shows the addition of the local frames into the point cloud.

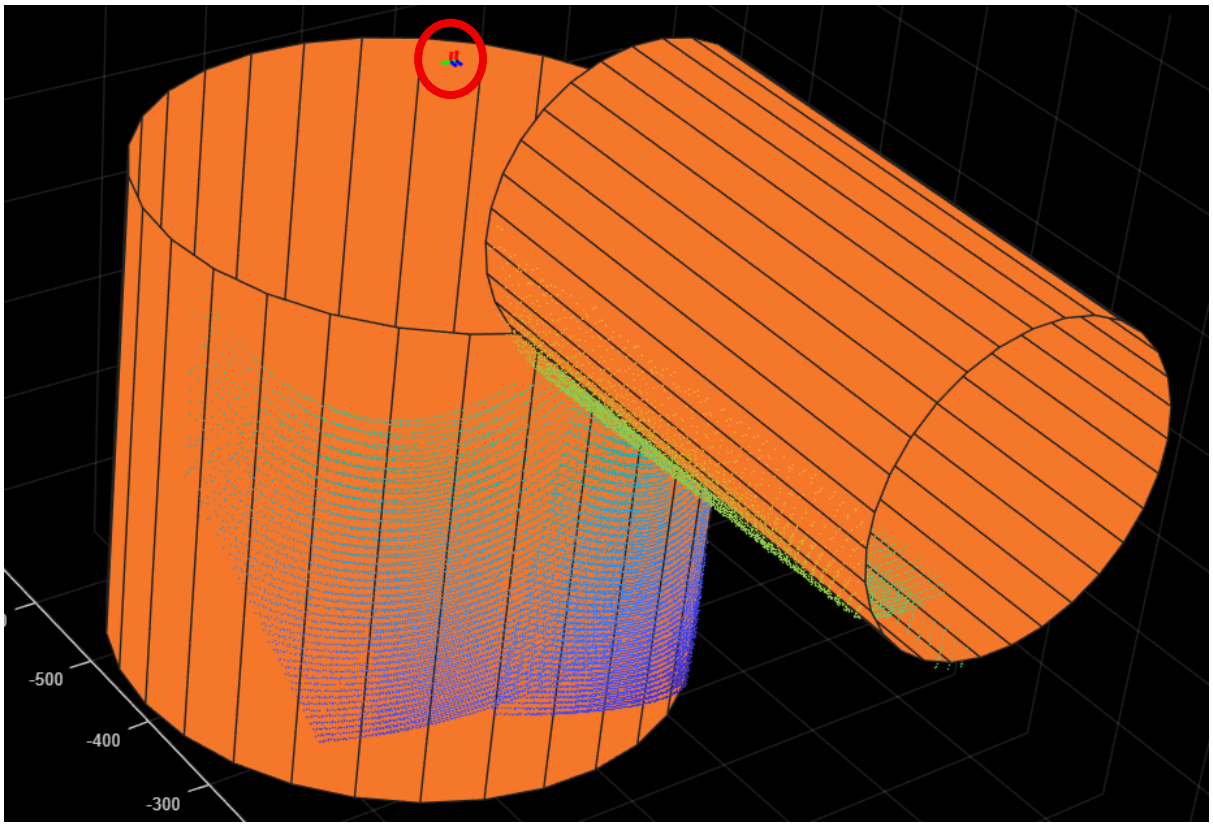


Figure 26 Fitted cylinders with local coordinate frames visualised

The previous equations were then used to generate the scan trajectory waypoints for scans 1 and 2 within local coordinate frames 1 and 2 respectively. Note that the waypoints in scan path 1 were all given a y offset equal to the y distance between the two frames. This accounts for the cylindrical axes not aligning, the effect of this is shown in figure 27. This feature assumes an accurate cylinder fit rather than the assumption of the pipes accurately aligning in real life and can be omitted if necessary.

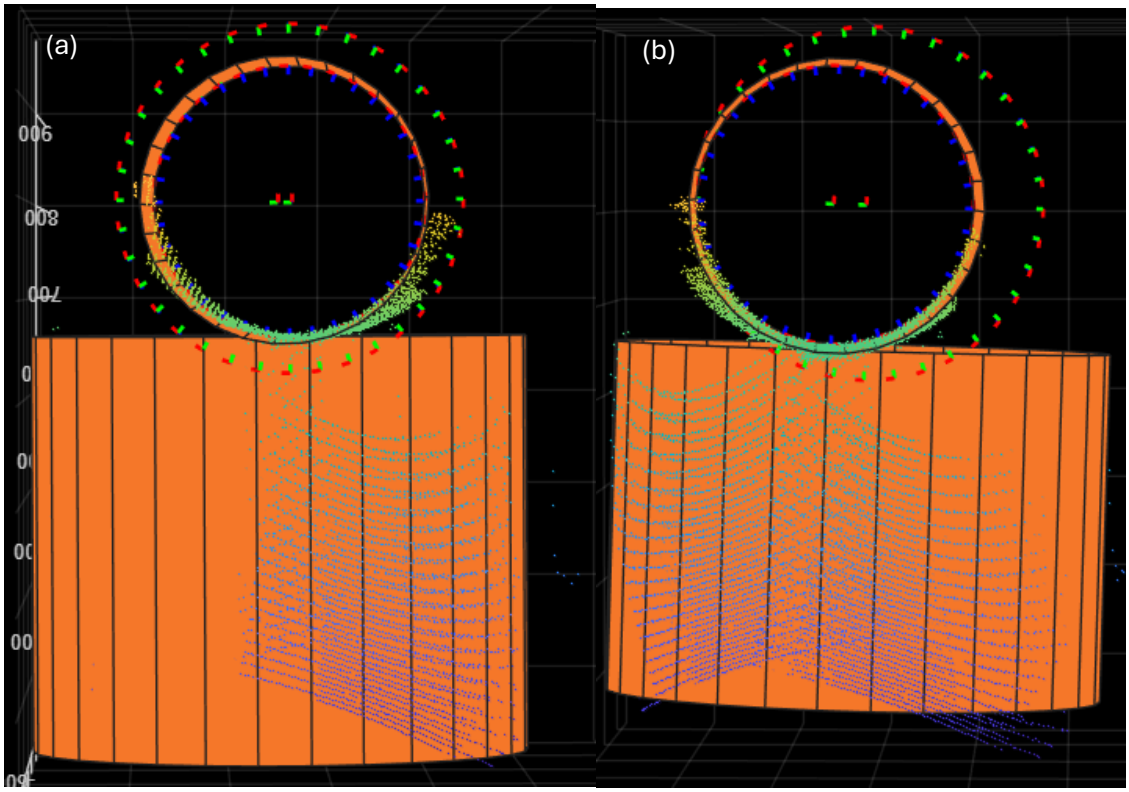


Figure 27 (a) Generated TRF path with offset included ensuring constant offset from fitted vertical cylinder. (b) offset omitted showing inaccurate positioning of scan path 1 for erroneous cylinder fit

In order to transform the waypoints from the local coordinate frame to the world frame, the rotation matrices and translation vectors between the local frames and the world frame were substituted into equation (1).

The ransac_and_orientation function within the Github repository performs the cylinder fits and determines the rotation and translation matrices for the local coordinate frames. These are input to the generate_waypoints and generate_waypoints_1 functions which output the scan trajectory waypoints with respect to the WRF for scans 1 and 2 respectively. The plot_waypoints function can be used to plot the waypoint coordinate frames if wanted. Figure 28 shows a point cloud with fitted cylinders and the path generated. Waypoints are spaced by 1 degree about the axis of the vertical pipe, resulting in 360 for each scan. This can be easily altered.

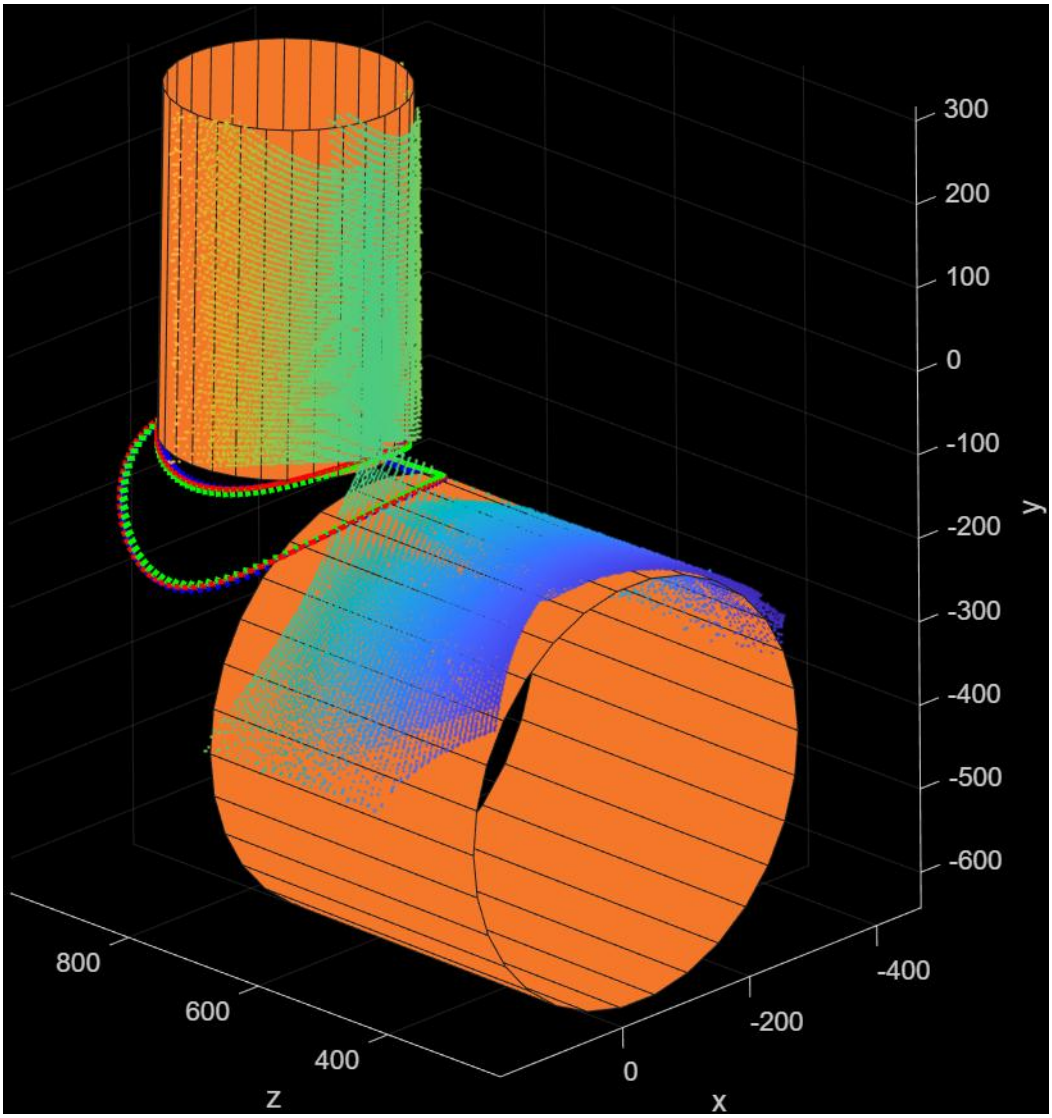


Figure 28 Point cloud with cylinders fitted and scan paths generated

3.2.3 Path execution

The position of the robot upon the pipe affects which waypoints are physically reachable, therefore a function (feasibility_checker) was created to derive a list of valid waypoints following a path generation. This function makes use of the robot error state and built in inverse kinematics of the Meca500. Upon a cartesian positional command, the Meca500 calculates the required joint configuration for the target pose. The robot enters its error state and produces an error message if no joint configuration would allow the desired target pose. The function on_robot_error was added to the class to handle infeasible positional commands; this function clears the error upon an infeasible positional command allowing the program to continue. The feasibility_checker function is comprised of the following steps:

1. Calculates the cartesian distance to each waypoint
2. Extracts the closest waypoint in cartesian space
3. Gives a MovePose command with the corresponding waypoint as the target pose, instantaneously cancelling the motion
4. The program increments the index of the target waypoint by 1 until an error is received due to an infeasible positional command. The error is handled by on_robot_error and the program continues

5. The same process is repeated with the target waypoint index being incremented by -1 until another error is received
6. The function keeps track of all waypoints which did not cause an error and stores them in a list of ascending order. The four edge waypoints are removed to avoid joint extremities

Having derived the feasible waypoints, the move_to_start function is executed which moves the robot end effector above the first waypoint with an offset of 10 mm. The end effector then moves towards the surface until contact is made i.e. a force of 6 N is observed on the force-torque sensor, which is the target force to be maintained throughout the scan. This instruction is contained within move_to_sample.

Once the target force has been reached, the end effector motion is governed by a PID controller which combines the target trajectory from the generated waypoints with force-torque feedback to ensure consistent coupling. Figure 29 summarises the PID control system.

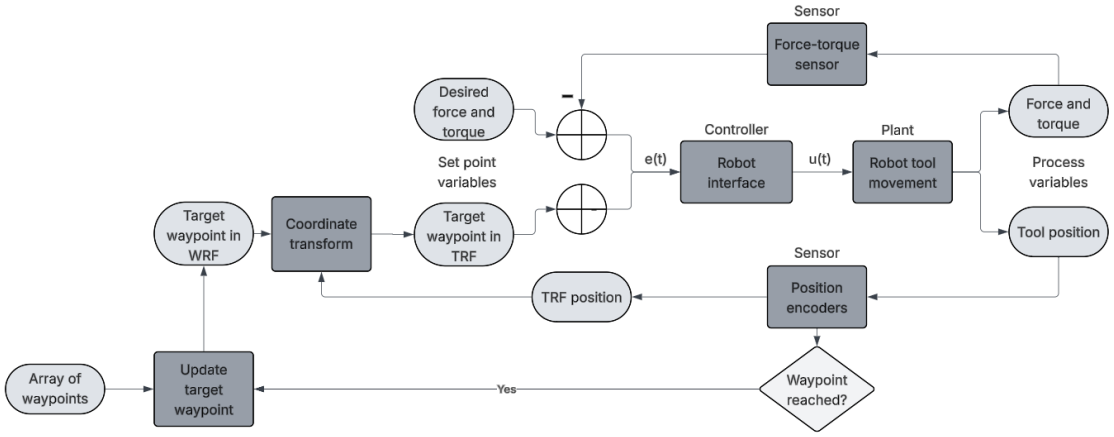


Figure 29 Flowchart for PID control system

Implementing this control system into code involved creating nine PID control functions (one attributed to each 3D translation and Euler angle, with three additional functions attributed to force-based corrections along the z axis and torque-based corrections about the x and y axes). A waypoint is considered reached when all translation and orientation descriptors of the end effector are within a certain tolerance of those of the waypoint. This is 1 mm for the x and y axes and 1 degree for the z axes. Tolerances for the z axis translation and x and y axes rotations are slightly higher to allow for the force torque corrections. The translational and rotational tool velocities are simply calculated by adding together the positional-based velocities and the correction velocities. Gain values were derived through experimental trial and error, discussed in the testing section.

This control system is contained within saddle_path and utilises the simple-pid library [59]. The program also contains code for plotting the force and torque readings over the duration of the scan as well as the TRF positions and orientations.

4 Testing

4.1 Laser scanning

The accuracy of the point cloud generation process was tested by conducting a rough scan on the wooden block shown in figure 30 (43 x 43 x 73 mm).



Figure 30 Wooden block for laser test

The generated point cloud displayed in MATLAB is shown in figure 31. As can be seen the dimensions of the front face roughly match the 43 mm which was expected.

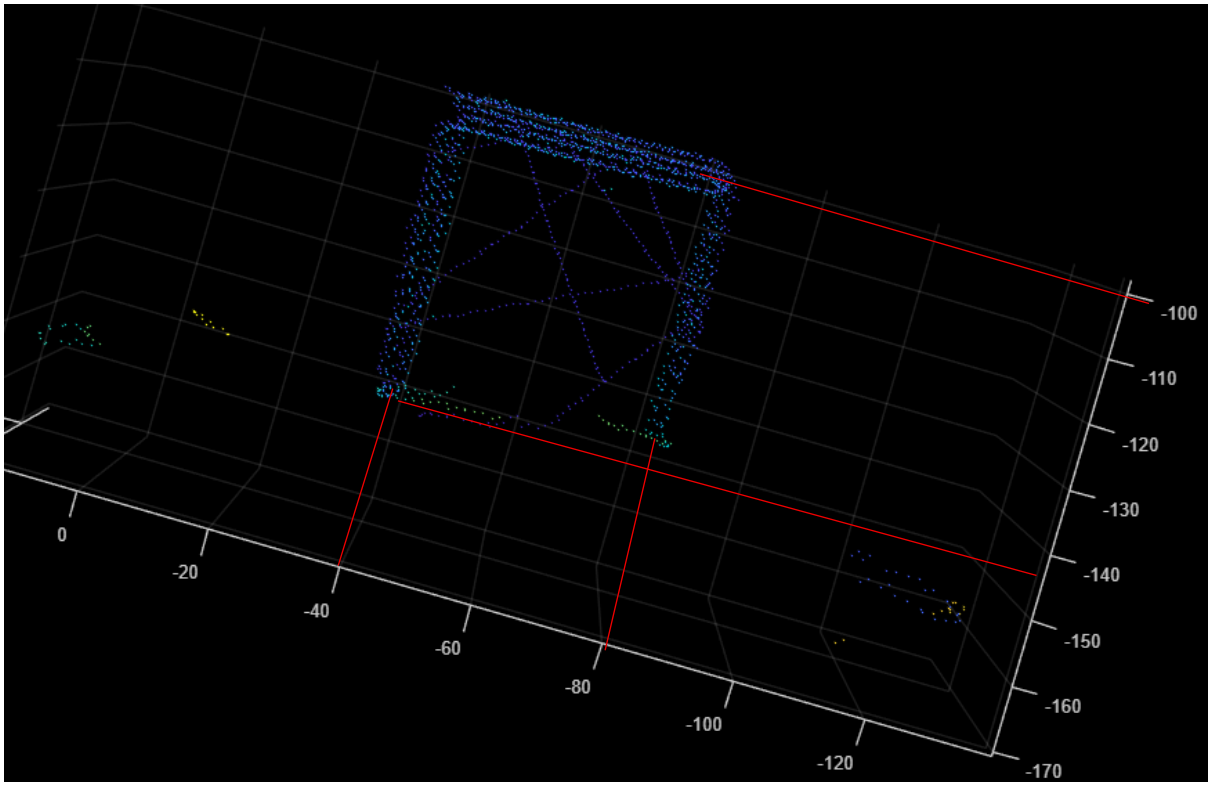


Figure 31 Generated point cloud from laser scan on wooden block showing approximate 43 x 43 mm front face

In order to ensure the reliability of the cylinder fit function, a test image was captured of the mock inspection sample using a Zivid light camera which produced a highly dense and accurate point cloud. The output 3D image with fitted cylinders is shown in figure 32. CloudCompare [60] is another software which was used to apply RANSAC, with the produced image shown in figure 33. MATLAB was chosen as the preferred software for the system due to its familiarity and compatibility with Python.

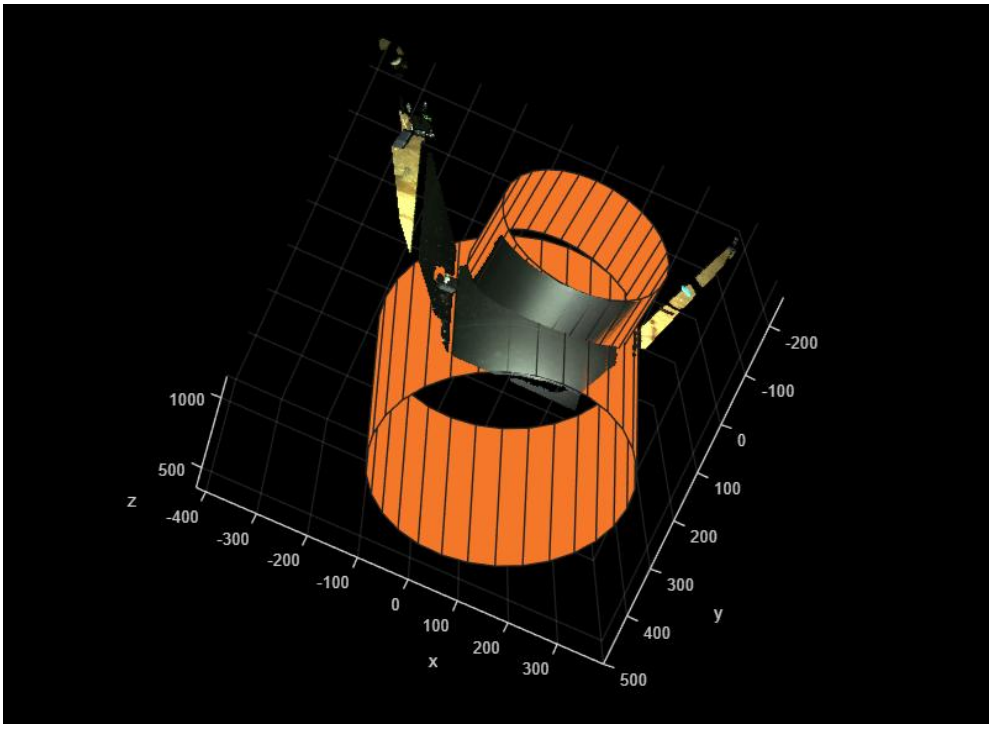


Figure 32 MATLAB cylinder fit on point cloud image generated using Zivid camera

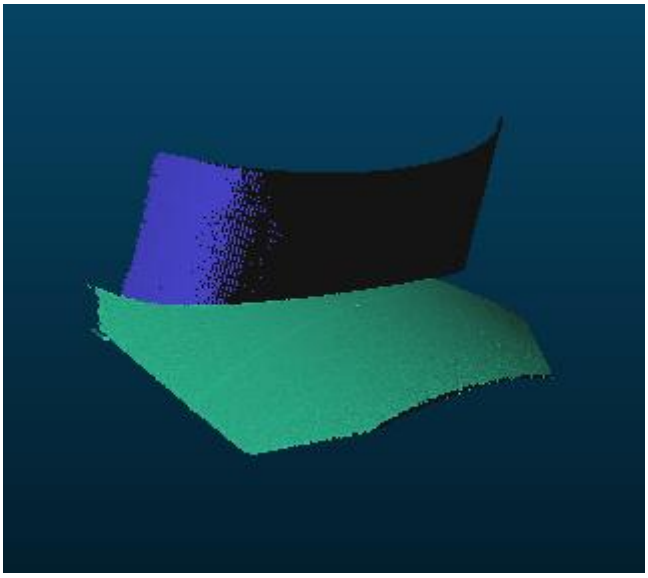


Figure 33 CloudCompare cylinder fit on point cloud image generated using Zivid camera

The speed and motion of the laser during point cloud data acquisition directly affects the density and overall accuracy of the produced point cloud. Laser scans were conducted at slow, medium and fast speeds and with several different motion paths to observe the

significance of these variables. This data can be used for time optimisation. Figure 34 illustrates the visual differences between point clouds produced using different speeds and motion paths.

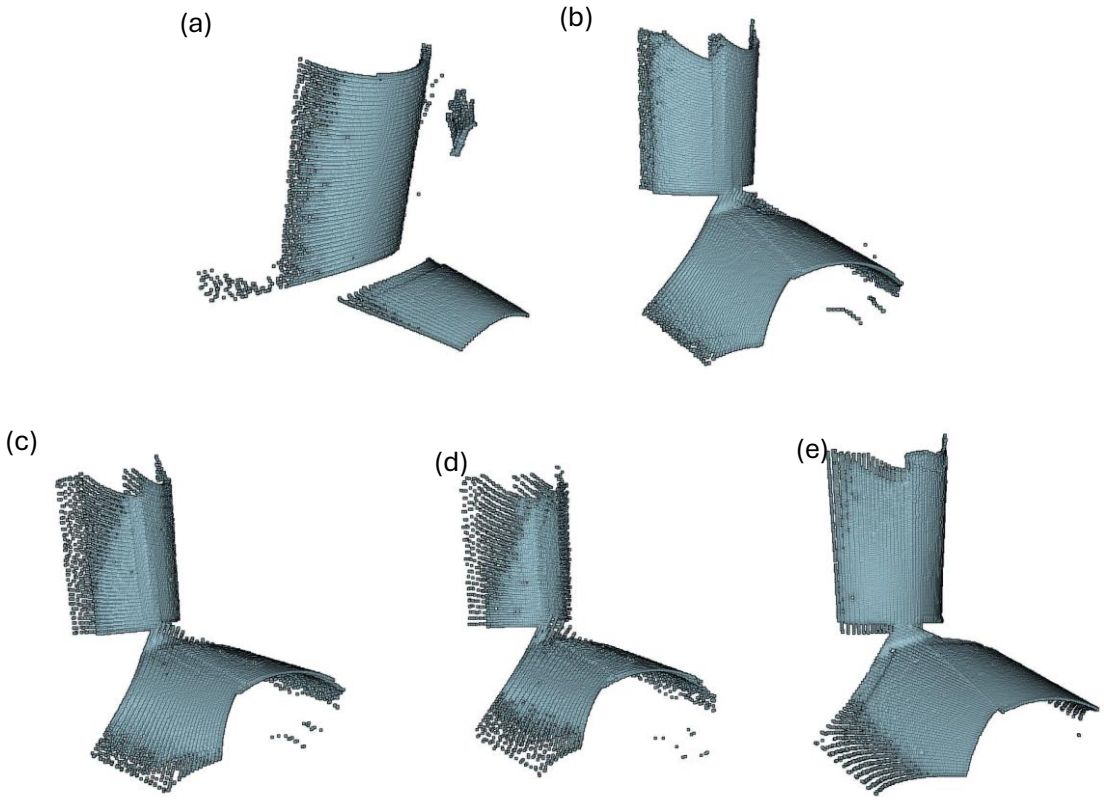


Figure 34 (a) two scans (base positioned centrally), slow speed (b) four scans (base positioned to the left, then to the right), slow (c) four scans, medium (d) four scans, fast (e) four scans, slow, axial rather than circumferential

The quality of the point cloud affects the resemblance of the fitted cylinders to the actual pipes. Small inaccuracies can be accounted for with the addition of force-torque feedback. All of the scans were conducted and RANSAC was applied several times to each of the point clouds and the pipe radius predictions, offset between local coordinate frames and time taken to find a best fit cylinder were observed. The results are shown in table 3, along with the approximate time taken for each scan.

Table 3 Results from laser scanning tests

Scan technique	Approx. time taken for scan (mins)	Time taken to fit cylinders (s)	Avg. local frame offset (mm)	Avg. error in radii (mm)
A	3.9	7.2 (often wrongly fitted cylinder)	100	150 (due to wrongly fitted cylinders)
B	9.3	7.3	10	5
C	4.1	5.7	15	10
D	2.3	5.6	30	15
E	9.3	7.7	30	30

The circumferential scans produced significantly better results than the axial scans. This likely relates to the way RANSAC works. For every ‘raster’ in a circumferential scan, a

subset of points following the profile of a cylinder are created. If the algorithm works in small local regions for each iteration, it is likely that several points from the same raster will be selected creating a consistent consensus. Because for the axial scan the robot repositions itself for every incremental step around the cylinder profile, the cylinder profile is less likely to be accurately followed.

4.2 Force-torque feedback

Appropriate proportional, integral and derivative gain terms had to established for the system to work optimally. This first involved experimentation with end effector PID control on a sample dome. The Ziegler-Nichols tuning method [61] was used to derive gain terms. Different responses from varying gain terms are shown in figure 35.

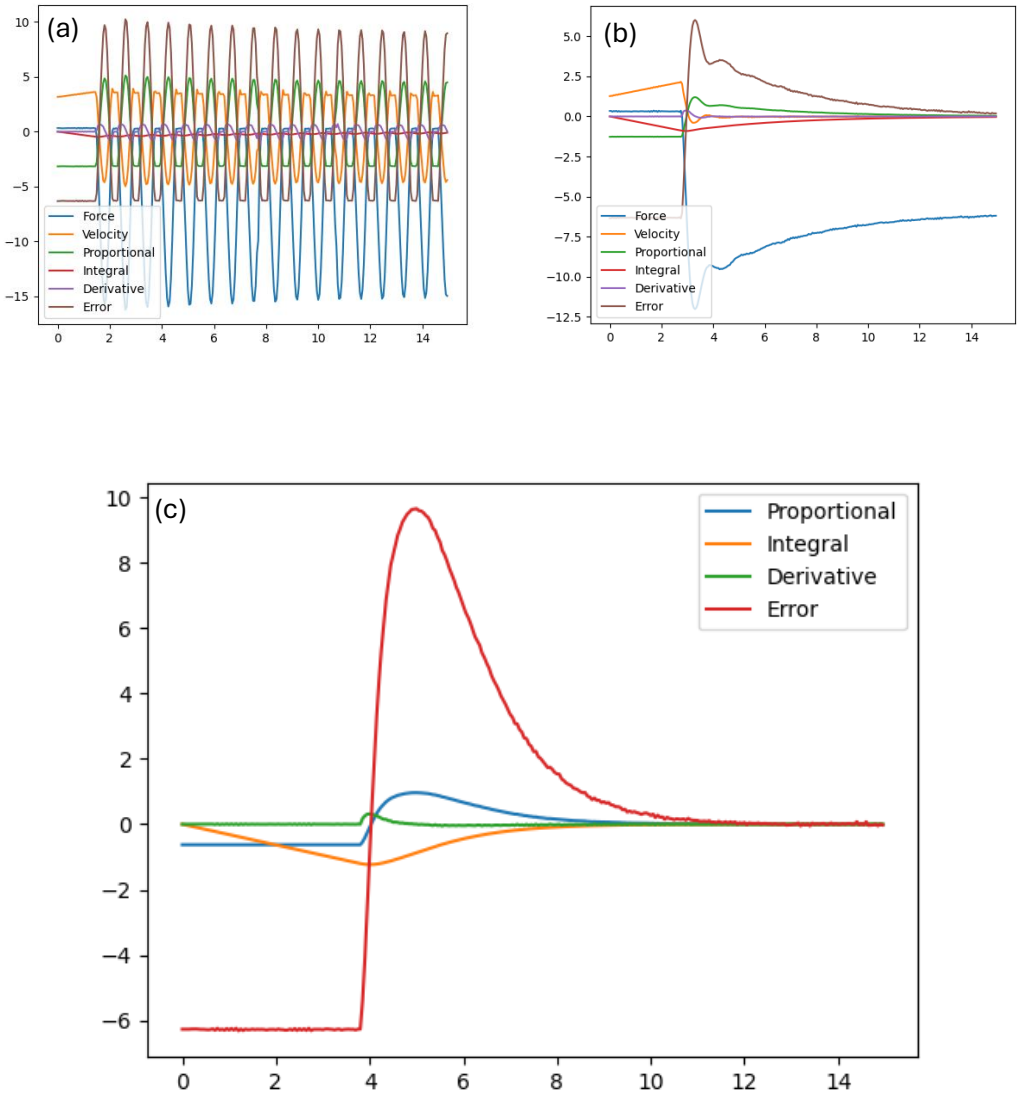


Figure 35 (a) system response with extreme gain causing significant oscillations (b) system response with moderate gain resulting in minor oscillations (c) system response with finely tuned gain showing convergence

Derivative gain was of little to no utility, generally causing additional oscillations, so this term was omitted from the control system

The mock inspection system was used to investigate the reliability of the PID control system for the saddle scan as well as the accuracy of the waypoints with respect to the desired scan path. The robot was positioned with a known offset from the local coordinate frame and the radii of the mock pipes were hard coded into the program. The gain terms were varied and force and torque readings from the sensor were plotted. Figure 36 shows the force and torque readings for the optimised gain terms. The force set point was -6 N and the torque set points were both zero. As can be seen the force remains relatively close to -6 N with variations of approximately 1 N. Small defects in the sample surface likely caused these minor oscillations. A gradual increasing offset can also be observed. This can perhaps be attributed to the changing surface gradient as the end effector moves to the top of the pipe's outer surface. The torque readings also exhibit minor oscillations and offsets, with the offset in the y torque decreasing over time. The offsets are most likely attributed to minor inaccuracy in the waypoint orientations. The decreasing y torque offset is probably due to the changing surface gradient, similar to the force case.

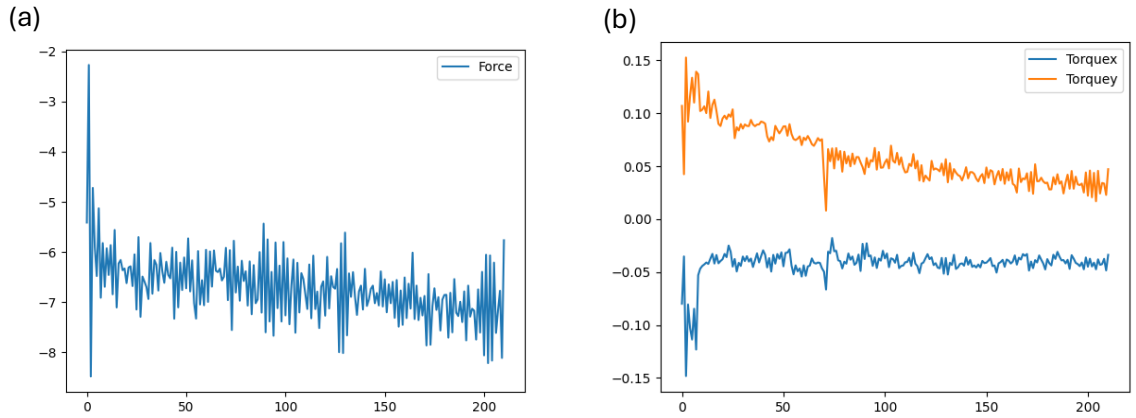


Figure 36 (a) force response from optimised PID parameters (b) torque responses from optimised PID parameters

Further testing was halted due to reliability issues with the force-torque sensor, so minimal data was collected within this area.

4.3 Overall system

The complete system was integrated together for testing. Again, due to reliability issues with the force-torque sensor, no actual ultrasonic data could be acquired, and only non-contact mock scans could be conducted. The optimised parameters were nonetheless implemented into the program, and the system produced a mock scan with an approximate accuracy of 5 – 10 mm from the desired path over the duration of the scan for a single test.

5 Further work

Having developed the basis of an autonomous saddle weld inspection system, there is considerable opportunity for improvement and further work. The following additions are some examples.

Machine learning

The system currently operates only on the dataset collected for the current scan. Machine learning algorithms could be implemented so that the path generation and execution algorithms become progressively more accurate as more tests are conducted. This would

involve incorporating past point cloud, force-torque and ultrasonic data as a means to optimise each scan. For example, the system could use past ultrasonic data to maintain a more accurate offset from the weld, as currently the horizontal offset remains at 40 mm, not accounting for the changing wedge angle. Weld tracking using ultrasonic data has been explored by the University of Strathclyde [39].

Machine learning algorithms could also be used as a means for responding to unexpected anomalies in a pipe surface.

Collision avoidance

A significant issue with the current system is its lack of spatial awareness. Often when repositioning the arm between laser and ultrasonic scans the arm is at risk of colliding with one of the pipes. Future work could involve implementing a collision avoidance algorithm which ensures that there is no risk of a collision throughout the entirety of the scanning process. This may involve additional visual feedback providing the robot with detailed awareness of both pipe locations. Having generated a point cloud, the robot could use this data to define a new workspace for repositioning the end effector which does not include any objects.

Scan time optimisation

Current laser scanning techniques have not been optimised and merely provide a basis for generating a point cloud with extensive coverage. All scans involve motion of only two joints and are conducted in separate stages by repositioning joints 1, 2 and 3 between stages. An optimised laser scan would involve rotation of all joints simultaneously and would ideally be planned in cartesian rather than joint space. The current scan path is also specific to the where the robot was positioned during testing and the size of the pipe sample. An improved system could involve a generalised scanning algorithm. This would require additional feedback before the laser scanning stage e.g. visual.

Full body planning

All tests were conducted on a stationary system situated on top of the pipe section. A complete system would utilise the roller to gather point cloud data from different locations about the pipe intersection. Point clouds generated from different locations could be combined to create a complete 3D description of the scene. Fusion of individual scenes to create a complete map of a robot's environment has been explored by the Oxford Robotics Institute [62].

Path planning for the improved system would include the trajectory of the roller as well as the robot end effector.

Singularity avoidance

A common issue which arose while conducting US scans was the arm running into singularities. This caused the arm to get stuck requiring to be reset. Attempts to resolve the issue included utilising the configuration commands so that the arm would begin scans in a 'safer' configuration which minimised joint values. This alleviated the issue to some degree but was very specific to the robot's location with respect to the scan path. Resetting the arm to its default position after a set number of waypoints was also attempting however this simply increased collision likelihood and slowed down the process significantly.

An algorithm to plan the entire path before any motion is undergone would be a significant improvement to the system. This algorithm could also incorporate blending, which would result in a smoother, more continuous trajectory.

6 Conclusion

Overall, the proposed system provides a strong foundation for a generalised robotic system for pipe inspection. Most of the initial objectives were met, hindered only by reliability issues with the force-torque sensor which eradicated the possibility of conducting any UT.

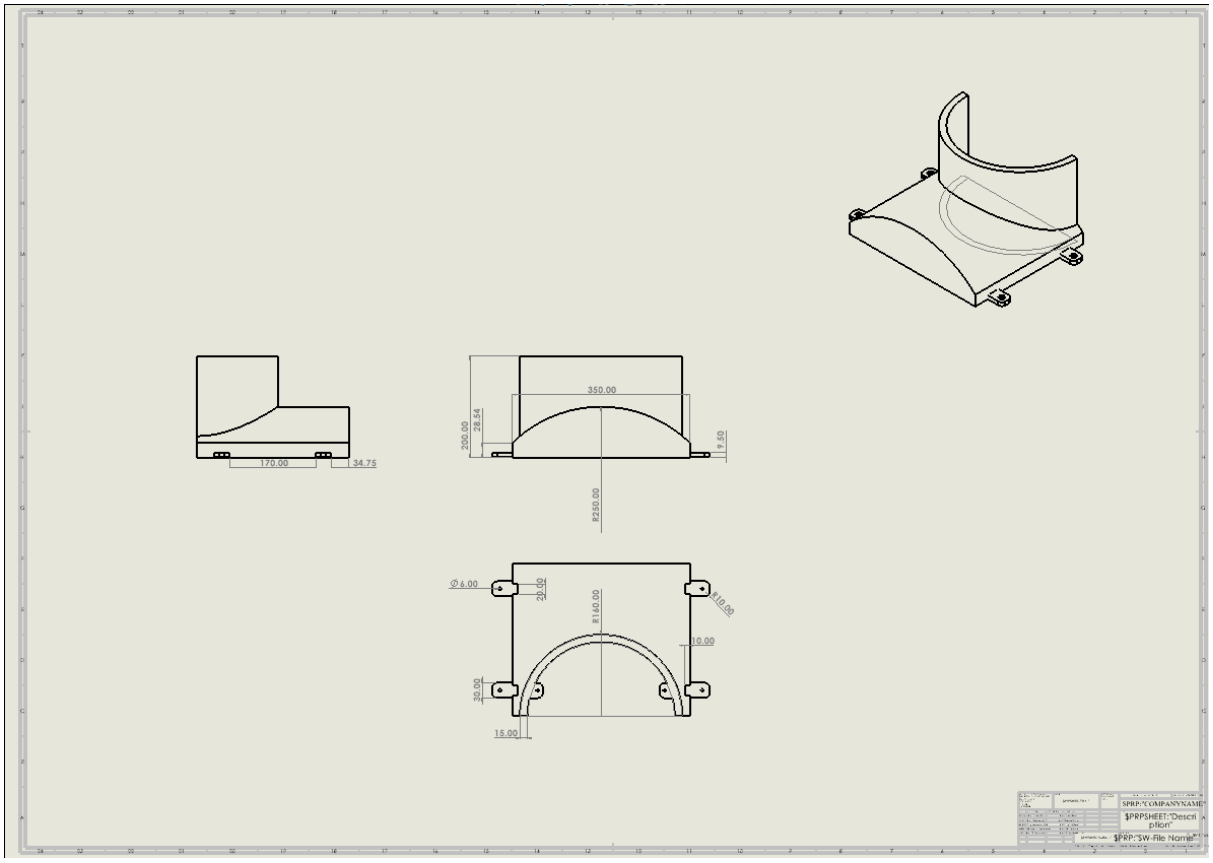
Extensive research into the field of automated NDT was conducted alongside practical experimentation in order to drive the body of work in the right direction. The resultant system encompasses a wide range of technical disciplines, such as scene recognition, path planning and PID control. It combines preliminary knowledge, such as the weld profile and general location of the pipe intersection, with collected data from a laser scanner and force-torque sensor in order to mathematically generate a path comprised of a series of waypoints and to execute motion between waypoints using PID control.

Ways to optimise the system were explored, such as experimentation with different laser scanning speeds and techniques to generate a point cloud, and variation of P, I and D gain terms for the control system. It was found that a slow scan covering a significant proportion of the pipe profiles was needed for the RANSAC cylinder fit and path generation to be accurate enough, and that minor variation in the force and torque with respect to their set points was inevitable. Although no UT was undergone, the system was tested by means of a non-contact mock scan and provided accuracy of approximately 5 – 10 mm.

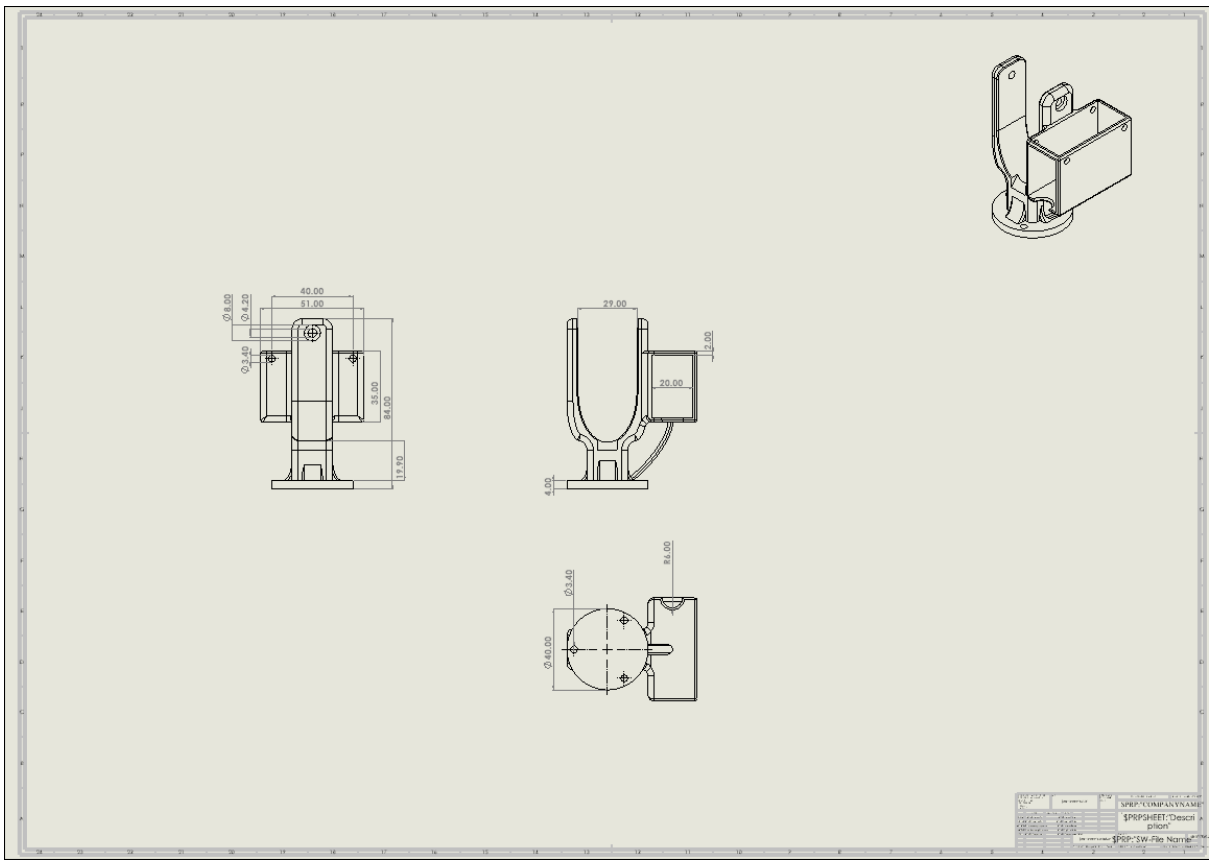
The body of work is open to a wide range of improvements such as machine learning algorithms, collision avoidance, scan time optimisation, full body planning and singularity avoidance.

7 Appendices

7.1 Appendix A: Mock pipe section 3D drawing



7.2 Appendix B: Mount 3D drawing



8 References

[1] C. Pengchao, “Advancements and future outlook of safety monitoring, inspection and assessment technologies for oil and gas pipeline networks,” *Journal of Pipeline Science and Engineering*, p. 100267, Mar. 2025, doi: 10.1016/j.jpse.2025.100267.

[2] A. R. Greig, “Automatic Inspection of Complex Geometry Welds,” University of London, London, 1992.

[3] C. Mineo and Y. Javadi, “Robotic Non-Destructive Testing,” Oct. 01, 2022, *MDPI*. doi: 10.3390/s22197654.

[4] Iowa State University, “https://www.nde-ed.org/NDETechniques/Ultrasonics/EquipmentTrans/erroranalysis.xhtml.”

[5] B. T. J. and M. T. Raj, *Practical non-destructive testing*. New Dehli: Woodhead Publishing, 2002.

[6] F. Abdollahi-Mamoudan, C. Ibarra-Castanedo, and X. P. V. Maldague, “Non-Destructive Testing and Evaluation of Hybrid and Advanced Structures: A Comprehensive Review of Methods, Applications, and Emerging Trends,” *Sensors*, vol. 25, no. 12, p. 3635, Jun. 2025, doi: 10.3390/s25123635.

[7] N. Meyendorf, N. Ida, R. Singh, and J. Vrana, “NDE 4.0: Progress, promise, and its role to industry 4.0,” *NDT and E International*, vol. 140, Dec. 2023, doi: 10.1016/j.ndteint.2023.102957.

[8] “NDE 4.0: The Fourth Revolution in Non- Destructive Evaluation: Digital Twin, Semantics, Interfaces, Networking, Feedback, New Markets and Integration into the Industrial Internet of Things”.

[9] S.-J. Song, J. Shin, and H. Jang, “Development of an ultra sonic phased array system for nondestructive tests of nuclear power plant components,” 2002. [Online]. Available: www.elsevier.com/locate/nucengdes

[10] R. Dahlstrom, “Robotic NDE for Industrial Field Inspections,” in *Handbook of Nondestructive Evaluation 4.0*, Springer International Publishing, 2021, pp. 1–22. doi: 10.1007/978-3-030-48200-8_17-1.

[11] H. Huang *et al.*, “Design and performance analysis of a tracked wall-climbing robot for ship inspection in shipbuilding,” *Ocean Engineering*, vol. 131, pp. 224–230, 2017, doi: 10.1016/j.oceaneng.2017.01.003.

[12] “Voliro.com.”

[13] U.S. Department of Labor, “Commercial Diving - Hazards and Solutions | Occupational Safety and Health Administration.” Accessed: Aug. 26, 2025. [Online]. Available: https://www.osha.gov/commercial-diving/hazards-solutions#:~:text=Commercial%20divers%20are%20exposed%20not%20only%20to%20the,from%20the%20operation%20of%20heavy%20equipment%20under%20water.

[14] *AUV 2014 : 2014 IEEE/OES Autonomous Underwater Vehicles (AUV) : October 6-9, Oxford, Mississippi*. IEEE, 2014.

[15] “3Robotics_Path_Planning”.

[16] Advanced Navigation, “https://www.advancednavigation.com/glossary/slam/#:~:text=Simultaneous%20Localization%20and%20Mapping%20(SLAM)%20is%20a%20technology%20used%20by,prior%20knowledge%20of%20their%20surroundings.”

[17] Y. Wang, H. Zhu, Y. Yu, and B. Hu, “The Path Planning and Location Method of Inspection Robot in a Large Storage Tank Bottom,” *Comput Intell Neurosci*, vol. 2023, no. 1, Jan. 2023, doi: 10.1155/2023/3029545.

[18] Mecademic, “Programming manual for Mecademic Robots (for firmware 10.2),” 2015.

[19] Hugh F. Durrant-Whyte, *Integration, Coordination and Control of Multi-Sensor Robot Systems*. Kluwer, 2012.

[20] Z. Zhou, Y. Zhang, and K. Tang, “Sweep scan path planning for efficient freeform surface inspection on five-axis CMM,” *CAD Computer Aided Design*, vol. 77, pp. 1–17, Aug. 2016, doi: 10.1016/j.cad.2016.03.003.

[21] Q. Zou and J. Zhao, “Iso-parametric tool path planning for point clouds,” 2013.

[22] M. Zhang, M. Sutcliffe, P. I. Nicholson, and Q. Yang, “Efficient Autonomous Path Planning for Ultrasonic Non-Destructive Testing: A Graph Theory and K-Dimensional Tree Optimisation Approach,” *Machines*, vol. 11, no. 12, Dec. 2023, doi: 10.3390/machines11121059.

[23] M. Tabatabaeipour *et al.*, “Mitigating RGB-D camera errors for robust ultrasonic inspections using a force-torque sensor,” *Nondestructive Testing and Evaluation*, 2024, doi: 10.1080/10589759.2024.2386618.

[24] R. Deepu, B. Honnaraju, and S. Murali, “Path generation for robot navigation using a single camera,” in *Procedia Computer Science*, Elsevier B.V., 2015, pp. 1425–1432. doi: 10.1016/j.procs.2015.02.061.

[25] R. Gr. Maev *et al.*, “NDE in the Automotive Sector,” in *Handbook of Nondestructive Evaluation 4.0*, Springer Nature Switzerland, 2024, pp. 1–38. doi: 10.1007/978-3-030-48200-8_21-2.

[26] A. Torres, M. Sílvia, E. João, P. T. Pereira, and L. M. Oliveira, “Advanced Structured Materials Additive Manufacturing Hybrid Processes for Composites Systems.” [Online]. Available: http://www.springer.com/series/8611

[27] E. Zhang, X. Shen, S. Wang, and S. Huang, “Normal calculation of artillery cradle before the robot arm assisted ultrasonic phased array automatic detection,” *Proc Inst Mech Eng C J Mech Eng Sci*, vol. 238, no. 6, pp. 2228–2236, Mar. 2024, doi: 10.1177/09544062231189705.

[28] J. Riise, L. Farr, M. Lindop, G. Pierce, P. I. Nicholson, and I. Cooper, “Methods for improving the accuracy of automated NDE systems.”

[29] London South Bank University, “https://www.lsbu.ac.uk.”

[30] T. P. Sattar and A. A. Brenner, “Robotic system for inspection of test objects with unknown geometry using NDT methods,” *Industrial Robot*, vol. 36, no. 4, pp. 340–343, 2009, doi: 10.1108/01439910910957093.

[31] Z. Fan, K. Bai, and C. Chen, “Ultrasonic testing in the field of engineering joining,” Jun. 01, 2024, *Springer Science and Business Media Deutschland GmbH*. doi: 10.1007/s00170-024-13569-w.

[32] TWI, “https://www.twi-global.com/media-and-events/press-releases/2018-01-twi-pioneers-invisible-weld-process.”

[33] M. Zhang, M. Sutcliffe, D. Carswell, and Q. Yang, “Robotic path planning using NDT ultrasonic data for autonomous inspection,” *International Journal of Metrology and Quality Engineering*, vol. 14, 2023, doi: 10.1051/ijmqe/2023013.

[34] “Mean Shift Clustering: A Comprehensive Guide.” Accessed: Aug. 12, 2025. [Online]. Available: https://www.datacamp.com/tutorial/mean-shift-clustering

[35] K. Liu, Z. Sheng, X. Yang, R. Xiao, and H. Chen, “Laser scanning–based robot welding path planning for complex saddle–shaped weld seams,” *Welding in the World*, vol. 69, no. 5, pp. 1385–1399, May 2025, doi: 10.1007/s40194-024-01881-8.

[36] Think Autonomous, “The Ultimate Guide to the RANSAC Algorithm.” Accessed: Aug. 12, 2025. [Online]. Available: https://www.thinkautonomous.ai/blog/ransac-algorithm/

[37] C. N. Macleod, “Considerations for Automated NDE Applications.”

[38] W. Jackson *et al.*, “QNDE2022-98358 MAGNETIC INSPECTION PLATFORM FOR TELEOPERATED REMOTE INSPECTIONS OF COMPLEX GEOMETRY.”

[39] A. Gilmour *et al.*, “Using Phased Array Ultrasound to Localize Probes During the Inspection of Welds,” *IEEE Open Journal of Instrumentation and Measurement*, vol. 2, pp. 1–10, Oct. 2023, doi: 10.1109/ojim.2023.3327484.

[40] A. Poole, M. Sutcliffe, G. Pierce, and A. Gachagan, “Autonomous, Digital-Twin Free Path Planning and Deployment for Robotic NDT: Introducing LPAS: Locate, Plan, Approach, Scan Using Low Cost Vision Sensors,” *Applied Sciences (Switzerland)*, vol. 12, no. 10, Jun. 2022, doi: 10.3390/app12105288.

[41] A. W. K. To, G. Paul, and D. Liu, “Surface-type classification using RGB-D,” *IEEE Transactions on Automation Science and Engineering*, vol. 11, no. 2, pp. 359–366, 2014, doi: 10.1109/TASE.2013.2286354.

[42] R. S. Dwyer-Joyce, “The application of ultrasonic NDT techniques in tribology,” Oct. 2005. doi: 10.1243/135065005X9763.

[43] Mecademic, “Programming manual for Mecademic Robots (for firmware 10.2),” 2015.

[44] “Compact & Precise Industrial Robots | Mecademic Industrial Robotics.” Accessed: Sep. 03, 2025. [Online]. Available: https://mecademic.com/

[45] Jireh Inspection Robotics, “NAVIC.” Accessed: Sep. 05, 2025. [Online]. Available: https://www.jireh.com/navic/

[46] ATI Industrial Automation, “Multi-Axis Force/Torque Sensor.” Accessed: Sep. 03, 2025. [Online]. Available: https://www.ati-ia.com/Library/documents/ATI_FT_Catalog.pdf

[47] Micro-Epsilon, “optoNCDT 1420.” Accessed: Sep. 03, 2025. [Online]. Available: https://www.micro-epsilon.com/fileadmin/download/manuals/man--optoNCDT-1420--en.pdf

[48] Eddyfi Electronics, “Phased Array Probes and Wedges.” Accessed: Sep. 05, 2025. [Online]. Available: https://www.eddyfi.com/en/product/phased-array-pa-probes-wedges

[49] Mecademic, “Mecademic/mecademicpy: Mecademic Python API.” Accessed: Sep. 04, 2025. [Online]. Available: https://github.com/Mecademic/mecademicpy

[50] “MEDAQLib.” Accessed: Sep. 03, 2025. [Online]. Available: https://www.micro-epsilon.co.uk/service/software-sensorintegration/medaqlib/

[51] “Source code for instruments.DSensor_ILD1420.” Accessed: Sep. 03, 2025. [Online]. Available: https://probercontrol.github.io/ProberControl/_modules/instruments/DSensor_ILD1420.html

[52] Matthew Brett, “transforms3d.” Accessed: Sep. 03, 2025. [Online]. Available: https://pypi.org/project/transforms3d/

[53] “PyVista — PyVista 0.46.3 documentation.” Accessed: Sep. 04, 2025. [Online]. Available: https://docs.pyvista.org/

[54] Wolfram Mathworld, “Steinmetz Curve -- from Wolfram MathWorld.” Accessed: Sep. 04, 2025. [Online]. Available: https://mathworld.wolfram.com/SteinmetzCurve.html

[55] MATLAB, “Computer Vision Toolbox.” Accessed: Sep. 04, 2025. [Online]. Available: https://uk.mathworks.com/help/releases/R2025a/vision/index.html

[56] MATLAB, “Call MATLAB from Python - MATLAB & Simulink.” Accessed: Sep. 04, 2025. [Online]. Available: https://uk.mathworks.com/help/matlab/matlab-engine-for-python.html

[57] MATLAB, “Robotics System Toolbox - MATLAB.” Accessed: Sep. 04, 2025. [Online]. Available: https://uk.mathworks.com/products/robotics.html

[58] Mathematics Stack Exchange, “Closest points between two lines.” Accessed: Sep. 05, 2025. [Online]. Available: https://math.stackexchange.com/questions/1993953/closest-points-between-two-lines

[59] Martin Lundberg, “simple-pid.” Accessed: Sep. 04, 2025. [Online]. Available: https://pypi.org/project/simple-pid/

[60] CloudCompare, “CloudCompare - Open Source project.” Accessed: Sep. 05, 2025. [Online]. Available: https://www.cloudcompare.org/

[61] Engineering LibreTexts, “9.3: PID Tuning via Classical Methods.” Accessed: Sep. 05, 2025. [Online]. Available: https://eng.libretexts.org/Bookshelves/Industrial_and_Systems_Engineering/Chemical_Process_Dynamics_and_Controls_(Woolf)/09%3A_Proportional-Integral-Derivative_(PID)_Control/9.03%3A_PID_Tuning_via_Classical_Methods

[62] M. Ramezani, G. Tinchev, and M. Fallon, “Online LiDAR-SLAM for Legged Robots with Robust Registration and Deep-Learned Loop Closure.”